Dominant Followership in AI-Driven Algorithmic Pricing

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Abstract

Pricing algorithms allow firms to monitor and respond to rival price changes at scale. This paper studies how such responsiveness affects competition when firms differ in demand-side market power. Using high-frequency cross-platform pricing data, I identify sequential price adjustments and measure firms' responses to rivals' earlier price changes. Amazon's behavior is distinct: rival changes predict its later adjustments, especially same-direction responses after rival increases. To investigate how this pattern affects prices, I develop a model of dominant followership. I find that a demand-advantaged follower may match rather than undercut rival price increases to monetize its existing customer base. This mechanism raises prices only when the demand-advantaged firm moves second. In Q-learning simulations, the same timing condition determines long-run outcomes: after observing rival price changes, the demand-advantaged firm learns response rules that sustain higher prices and reproduce Amazon's tendency to follow rival increases without pricing above them.

***Keywords*— Algorithmic Pricing, Demand-Side Market Power, Sequential Price Adjustment, Reinforcement Learning**

1 Introduction

Recent antitrust allegations highlight the strategic importance of price responses in algorithmic pricing. In its 2023 complaint against Amazon, the U.S. Federal Trade Commission (FTC) alleged that Amazon's internal pricing algorithm, "Project Nessie," used information about competitors' pricing responses to increase Amazon's profits by more than $1 billion (Federal Trade Commission 2024). The allegation illustrates a broader possibility: firms may use pricing algorithms to learn rivals' response patterns, potentially softening competition.

This possibility is especially consequential when the responding firm has substantial demand-side market power. Under the online-superstore market definition used in the FTC complaint, Amazon allegedly accounted for more than 82 percent of gross merchandise value among major online superstore retailers in 2022 (Federal Trade Commission 2024). This market position may reflect switching frictions, platform convenience, or other non-price advantages. Because these advantages allow Amazon to retain consumers without undercutting rivals, they change its incentives when responding to rival price changes. Rivals that learn these responses may condition their own prices on those responses, with consequences for market-wide outcomes.

These strategic effects may become more important as AI makes response-based pricing more scalable. Recent industry evidence suggests that firms increasingly expect generative and agentic AI to be used in pricing, with systems capable of monitoring rival prices and implementing systematic responses rapidly.[1] When price responses are implemented rapidly at scale, they may alter rivals' pricing incentives and thereby affect market-wide prices and consumer welfare.

This paper focuses on one channel through which such responses can matter: a dominant firm may retain more consumers than its rivals at comparable prices. The key economic force is that a price cut trades off additional consumers against reducing margins on existing consumers. Undercutting offers a larger demand gain to a smaller rival with few existing consumers. A dominant firm, by contrast, sacrifices margin on consumers it would otherwise retain and may prefer to match a rival's price increase. Anticipating this response, the smaller rival can set a higher price than it could against a follower with weaker demand. Sequential adjustment can therefore soften

[1] McKinsey's 2025 Agentic AI in Pricing Survey reports rapid expected adoption of generative and agentic AI in B2B pricing over the next one to three years; see McKinsey, "B2B pricing: Navigating the next phase of the AI revolution," April 7, 2026.

competition market-wide when the dominant firm follows.

I develop a framework to study this mechanism in AI-driven price competition. The framework isolates two dimensions of asymmetry: demand asymmetry, which governs how consumers are allocated across firms at different relative prices, and the timing of price adjustment, which determines whether a firm moves first or responds after observing a rival's posted price. I first characterize static pricing benchmarks to show how these dimensions shape incentives to undercut or match prices absent learning. I then introduce reinforcement-learning pricing algorithms in a sequential environment that captures the asynchronous nature of online retail: firms observe rivals' posted prices and respond rapidly but not instantaneously, earning flow demand before the next adjustment. By departing from the canonical simultaneous-move framework with symmetric firms (Calvano et al. 2020), this setting shows how learning evolves when firms differ in demand advantage and move sequentially.

The static benchmarks show that the identity of the follower matters for equilibrium prices. Prices are higher when the dominant firm follows than when the smaller rival follows. This price effect grows with the dominant firm's demand advantage: greater loyalty makes matching more attractive for the dominant follower, allowing the smaller rival to set higher prices when it moves first. The same force also aligns firms' role preferences. In the relevant price region, both firms prefer the smaller rival to move first and the dominant firm to follow because this configuration supports higher equilibrium prices. Demand asymmetry can therefore make dominant followership profitable for both firms, helping sustain a role configuration that raises prices and thereby reduces consumer welfare.

The reinforcement-learning simulations show that demand asymmetry affects prices primarily through its interaction with timing. Demand asymmetry alone does not substantially increase prices when firms adjust simultaneously. Once price adjustment is sequential, however, prices rise when the dominant firm follows: learning shifts toward high-price states in which matching is profitable for the dominant firm but not for the smaller rival. This configuration also significantly increases joint firm profits, making dominant followership sustainable from both firms' perspective and aligning firms' role preferences, even as consumers face higher prices and lower welfare.

I then construct a high-frequency panel of prices for identical products across major online platforms over a one-year period. The empirical analysis shows that Amazon combines several features central to the mechanism: persistent price competitiveness, frequent small and asynchronous price adjustments, and strong responsiveness to rival price changes. Amazon's lowest price position is often reached through matching, and its responses are especially pronounced following rival price increases. These patterns are consistent with a sequential pricing environment in which a dominant firm acts as a responsive follower.

The results show that algorithmic pricing can amplify market power by changing the strategic value of responding to rivals. When consumers are asymmetrically attached to firms, responsiveness itself becomes a source of advantage: the dominant firm can benefit from following, smaller rivals can benefit from leading, and consumers face higher prices supported by this role alignment. The paper therefore identifies algorithmic responsiveness as a mechanism that can amplify demand-side market power while making the resulting pricing roles attractive to asymmetric firms.

Related Literature. This paper contributes to the empirical literature on algorithmic pricing. Existing empirical work primarily studies whether the adoption or use of pricing algorithms affects price levels or margins in particular markets, including gasoline, housing, and online marketplaces (Assad et al. 2024; Calder-Wang and Kim 2023; Musolff 2024). Less is known about how algorithmic pricing operates through responsiveness: whether firms react to rivals' price changes within a given response window, and how these reactions vary with firms' market dominance. I document these response patterns directly in online retail, focusing on algorithmic responsiveness as a channel through which pricing technology can shape market-wide outcomes.

This paper also contributes to the empirical literature on e-commerce price competition. Existing work documents price dispersion, Buy Box competition, and algorithmic repricing within online marketplaces, especially Amazon Marketplace (Chen et al. 2016; He et al. 2022; Lam 2023; Hanspach et al. 2024; Musolff 2024). The Project Nessie allegations point to a different object of study: cross-platform pricing responses, in which one online retailer's price changes may affect pricing decisions by other retailers. Studying this setting is difficult because Amazon's prices are relatively visible, while rival retailers' prices are harder to observe at high frequency. I address this gap by constructing synchronized high-frequency pricing data that track identical products across major online retailers. This allows me to measure how price changes propagate sequentially across retailers.

The paper adds to the theoretical literature on algorithmic pricing. Canonical studies show that independently learning algorithms can generate supracompetitive outcomes in simultaneous-move pricing environments (Calvano et al. 2020; Banchio and Mantegazza 2023; Asker et al. 2024). These settings are less suited to markets in which posted prices are observable and firms can react to rivals' posted prices at different times, with demand realized between price adjustments. This paper shifts attention from algorithmic collusion in simultaneous or symmetric environments to systematic algorithmic responsiveness under asynchronous price adjustment.

Related work relaxes symmetry on the supply side, showing that speed-based commitment by a faster algorithmic firm can raise prices against a slower or myopic rival (Brown and MacKay 2025). Existing algorithmic-pricing models, however, still largely abstract from demand-side asymmetries, which are common in markets where loyalty, inertia, or other non-price advantages make one firm's pricing behavior especially consequential for rivals' subsequent pricing decisions. I also allow timing to matter, but study this distinct source of asymmetry in a learning environment. Rather than focusing on commitment by one faster firm, I study how learning algorithms adjust in a sequential environment where the value of responding depends not only on when firms move, but also on how demand is allocated when they match or undercut rivals' prices.

The paper also contributes to work on sequential price competition and price leadership. Maskin and Tirole (1988) provide a canonical model of alternating-move price competition, in which firms condition their pricing decisions on rivals' current prices. Klein (2021) extends this sequential pricing framework to Q-learning agents. A related empirical literature studies price leadership, showing that dominant firms can use leadership and price experiments to create focal points and coordinate market prices (Byrne and de Roos 2019). In contrast to this emphasis on dominant-firm leadership, the empirical patterns in this paper motivate a focus on dominant-firm followership: Amazon frequently responds to rivals' price changes. To capture this pattern in a learning framework, I extend sequential Q-learning models by introducing empirically motivated demand and timing asymmetries. The resulting framework shows that, when the follower has demand-side market power, moving second can be more valuable than leading.

Finally, the paper connects algorithmic pricing to the literature on demand-side market power. Search frictions, product differentiation, loyalty, and switching costs can give firms persistent demand advantages and soften price competition (Hotelling 1929; Diamond 1971; Varian 1980;

Narasimhan 1988; Beggs and Klemperer 1992). This literature primarily studies how demand-side frictions affect price levels, markups, market shares, or price dispersion. I show that these same frictions can also shape pricing roles in algorithmic markets: when firms differ in market dominance, demand-side advantages can align role preferences, making dominant followership and smaller-rival leadership jointly profitable.

2 A Model of Dominant Followership

This section develops a stylized model of dominant followership. The baseline model isolates the core mechanism in one-shot simultaneous and Stackelberg games, both of which clear the market once after all prices have been chosen. I then introduce flow demand in a three-revision alternating game and consider a loyal-switcher extension. Together, these exercises show how demand-side advantage can make followership strategically valuable.

2.1 Environment

There are two firms, indexed by $i \in \{D, S\}$, where D denotes the dominant firm and S the smaller rival. Firms choose prices from a finite evenly spaced grid

$$\mathcal{P} = \{\delta, 2\delta, \ldots, M\delta\},$$

where $\delta > 0$ is the grid step and $M\delta$ is the maximum price. Let $p_i \in \mathcal{P}$ denote firm i's price.

This discreteness can be interpreted as a reduced-form representation of limited consumer price perception: under Weber's law, the smallest noticeable change in a stimulus is proportional to its initial level, so sufficiently small relative price differences may generate the same perceived price and demand response (Fechner 1860; Monroe 1971, 1973).[2] In addition, firms may face operational or managerial costs of changing prices, so observed prices need not adjust continuously. Thus, the model's discrete price set can be viewed as a set of perceptually distinct price categories. Consistent with this interpretation, Appendix Figure A.3 shows that observed price adjustments are not infinitesimal: median dollar adjustment sizes rise with product price levels, while median

[2] Formally, if $\kappa > 0$ denotes the minimum relative price difference consumers perceive, then prices p and p' are perceived as distinct only if $|p - p'|/p \geq \kappa$. Prices with $|p - p'|/p < \kappa$ fall into the same perceived-price category.

relative adjustments remain economically meaningful across price groups.

Total demand is normalized to one, so the model focuses on how demand is allocated across firms rather than on changes in aggregate market size. Demand asymmetry is captured by a parameter $\lambda \in [1/2, 1]$, which determines the dominant firm's demand share when prices are equal:

$$q_D(p_D, p_S) = \begin{cases} 1, & p_D < p_S, \\ \lambda, & p_D = p_S, \\ 0, & p_D > p_S, \end{cases} \qquad q_S(p_D, p_S) = 1 - q_D(p_D, p_S).$$

The low-price firm captures the market, while equal prices split demand asymmetrically in favor of the dominant firm. The case $\lambda = 1/2$ is the symmetric benchmark; $\lambda > 1/2$ captures demand-side advantage at price parity.

Timing and market clearing. The baseline analysis compares two one-shot timing protocols. In the simultaneous game, firms choose prices without observing the rival's current choice. In the sequential, or Stackelberg, game, a leader chooses first and a follower observes the leader's price before choosing its own. Under both protocols, the market clears once, after both prices have been chosen. There is therefore no demand or payoff between the leader's action and the follower's response. With marginal cost normalized to zero, realized profit is

$$\pi_i(p_D, p_S) = p_i q_i(p_D, p_S).$$

The protocols differ only in whether one firm observes the other's price before acting.

2.2 Matching versus Undercutting

The key mechanism operates through the net payoff from undercutting. Consider a firm facing a rival price $p \in \mathcal{P}$. The firm can either match the rival's price or undercut it by one grid step. Undercutting creates two opposing effects. First, it increases demand: by charging $p - \delta$, the firm captures consumers who would otherwise buy from the rival at a price tie. Second, it lowers margins: the firm earns $p - \delta$, rather than p, on the consumers it would have served by matching.

These two effects depend on the firm's tie-demand share. If the firm's share at a price tie is s, matching yields sp, while undercutting yields $p - \delta$. The firm is therefore willing to match if and only if

$$sp \geq p - \delta,$$

or equivalently,

$$p \leq \frac{\delta}{1 - s}. \tag{1}$$

In the baseline model, $s = \lambda$ for the dominant firm and $s = 1 - \lambda$ for the small firm. Thus, a larger tie-demand share expands the set of rival prices the firm is willing to match.

Under the symmetric benchmark, $\lambda = 1/2$, this tradeoff is identical across firms. Each firm serves half the market at equal prices, so each firm gains the same additional demand from undercutting and sacrifices the same margin on its existing demand.

Under demand asymmetry, $\lambda > 1/2$, the tradeoff differs across firms. At a price tie, the dominant firm serves a larger share of the marginal market, while the small firm serves a smaller share. The small firm therefore has more incremental demand to gain from undercutting and less inframarginal demand on which it sacrifices margin. The dominant firm faces the opposite incentives: it already serves a larger share at parity, so undercutting adds less incremental demand while lowering the price on a larger existing customer base. As a result, the same rival price can make matching attractive for the dominant firm but undercutting attractive for the small firm.

I next apply this condition to simultaneous and sequential pricing.

2.3 Simultaneous Pricing Benchmark

Under simultaneous pricing, firms choose prices without observing the rival's current choice. Consider a candidate common-price equilibrium $p_D = p_S = p$. At this price, the dominant firm earns λp, while the small firm earns $(1 - \lambda)p$. A one-step undercut yields $p - \delta$ and captures the entire market. Undercutting by more than one grid step is never more profitable, since it also captures the entire market but at a lower price.

At a common price, the small firm is the more aggressive potential undercutter. It has less to lose from cutting price, because it serves a smaller share at parity, and more to gain, because a cut captures the market. Applying condition (1), its no-undercutting condition is

$$p \leq \frac{\delta}{\lambda}.$$

The dominant firm's no-undercutting condition is less restrictive:

$$p \leq \frac{\delta}{1-\lambda}.$$

Since $\lambda > 1/2$, we have

$$\frac{\delta}{\lambda} < \frac{\delta}{1-\lambda},$$

and the small firm's incentive constraint binds. The set of common prices satisfying both no-undercutting constraints is therefore

$$\left\{ p \in \mathcal{P} : p \leq \frac{\delta}{\lambda} \right\}.$$

Thus, the highest common-price equilibrium is

$$p^{Sim}(\lambda) = \max \left\{ p \in \mathcal{P} : p \leq \frac{\delta}{\lambda} \right\}.$$

The simultaneous benchmark therefore does not translate the dominant firm's weaker undercutting incentive into a higher common price: both firms' constraints must hold, and the small firm's stronger incentive to undercut pins down the outcome. In the symmetric benchmark, $\lambda = 1/2$, the highest common-price equilibrium is

$$p^{Sim}(1/2) = \max \left\{ p \in \mathcal{P} : p \leq 2\delta \right\}.$$

Since $\delta/\lambda < 2\delta$ when $\lambda > 1/2$, demand asymmetry weakly lowers the highest simultaneous common-price equilibrium.

2.4 Stackelberg Pricing and Follower Identity

In the Stackelberg game, the leader posts a price, the follower observes it and responds, and the market then clears once at the resulting price pair. This timing changes which constraint matters. The leader moves first, but the price it can sustain is disciplined by the firm that moves second. Anticipating the follower's response, the leader can set a high price only if that follower is willing to match rather than undercut. Thus, follower identity determines the highest price that can be sustained by matching.

Let p_D^{follow} denote the highest matched price sustainable when the dominant firm follows, and let p_S^{follow} denote the corresponding price when the small firm follows.

Proposition 1 (Dominant followership raises sustainable prices)**.** *Suppose $\lambda > 1/2$. The highest matched price sustainable under sequential pricing satisfies*

$$p_D^{follow} \geq p_S^{follow}.$$

Specifically,

$$p_D^{follow} = \max\left\{p \in \mathcal{P} : p \leq \frac{\delta}{1-\lambda}\right\},$$

while

$$p_S^{follow} = \max\left\{p \in \mathcal{P} : p \leq \frac{\delta}{\lambda}\right\}.$$

Proof. The leader's payoff is increasing in any price the follower matches. Hence, the highest sustainable matched price is the maximum grid price satisfying the relevant follower constraint. By condition (1), a dominant follower is willing to match a leader price p if and only if

$$p \leq \frac{\delta}{1-\lambda},$$

while the small follower is willing to match if and only if

$$p \leq \frac{\delta}{\lambda}.$$

If the dominant firm follows, this yields

$$p_D^{follow} = \max\left\{p \in \mathcal{P} : p \leq \frac{\delta}{1-\lambda}\right\}.$$

If the small firm follows, this yields

$$p_S^{follow} = \max\left\{p \in \mathcal{P} : p \leq \frac{\delta}{\lambda}\right\}.$$

Since $\lambda > 1/2$,

$$\frac{\delta}{1-\lambda} > \frac{\delta}{\lambda}.$$

Hence,

$$\left\{p \in \mathcal{P} : p \leq \frac{\delta}{\lambda}\right\} \subseteq \left\{p \in \mathcal{P} : p \leq \frac{\delta}{1-\lambda}\right\}.$$

The feasible set of matched leader prices is therefore weakly larger when the dominant firm follows. Since p_D^{follow} and p_S^{follow} are the maximum elements of their respective feasible sets,

$$p_D^{follow} \geq p_S^{follow}.$$

$\square$

Thus, follower identity changes sustainable prices by changing which undercutting constraint binds. A dominant follower relaxes the leader's constraint, while a small follower tightens it.

The inequality is weak because prices are chosen from a discrete grid; it is strict whenever $\mathcal{P}$ contains a price in

$$\left(\frac{\delta}{\lambda}, \frac{\delta}{1-\lambda}\right].$$

In the symmetric benchmark, $\lambda = 1/2$, the sustainable matched price is independent of follower identity and equals

$$p_{\text{sym}}^{Seq} = \max\left\{p \in \mathcal{P} : p \leq 2\delta\right\}.$$

For $\lambda > 1/2$, the cutoff for a dominant follower is above 2δ, while the cutoff for a small follower is below 2δ. Since each sustainable matched price is the highest grid point below its relevant cutoff,

$$p_D^{follow} \geq p_{\text{sym}}^{Seq} \geq p_S^{follow}.$$

Thus, unlike the simultaneous benchmark, where asymmetry is disciplined by the small firm's no-undercutting constraint, sequential pricing allows the same demand asymmetry to raise prices when the dominant firm moves second.

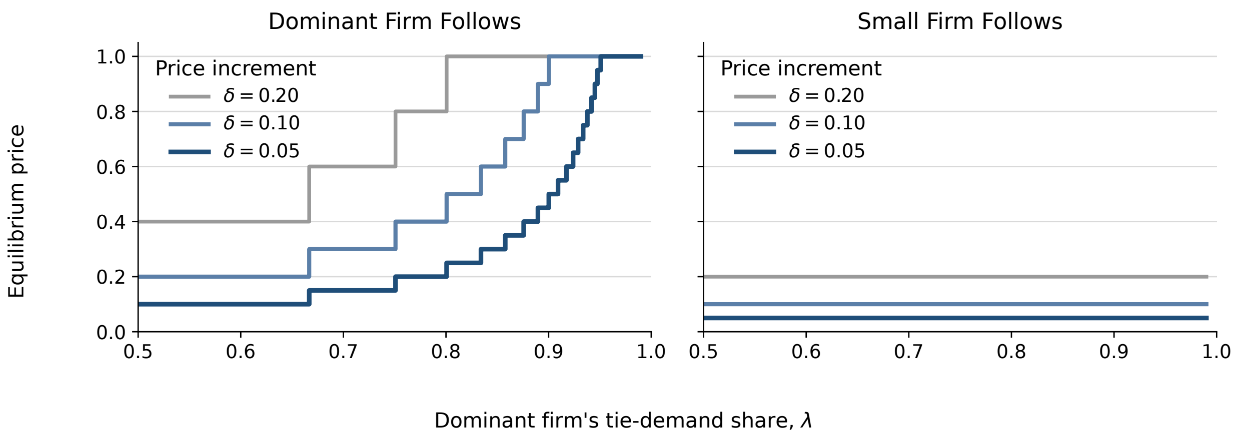

Figure 1. Sustainable Matched Prices under Sequential Pricing. The left panel shows the case in which the dominant firm follows, while the right panel shows the case in which the small firm follows. Prices are the highest grid points below the relevant matching cutoff.

Figure 1 illustrates that the price effect of demand asymmetry depends on follower identity. When the dominant firm follows, a larger λ weakens the follower's incentive to undercut, allowing the leader to sustain progressively higher matched prices. When the small firm follows, the same increase in λ strengthens the follower's incentive to undercut, keeping sustainable matched prices low. Thus, demand advantage raises prices only when the advantaged firm moves second.

2.5 Role Preferences

The ranking of sustainable matched prices has a direct implication for firms' preferred roles.

Payoff implication. Let $S \to D$ denote the matched sequential outcome in which the small firm leads and the dominant firm follows, and let $D \to S$ denote the reversed matched outcome. At any matched price, the dominant firm receives share λ and the small firm receives share $1-\lambda$, regardless of which firm led. Therefore, conditional on matched pricing, role preferences are determined only by the matched price.

Under $S \to D$, the matched price is p_D^{follow}. Under $D \to S$, the matched price is p_S^{follow}. Since Proposition 1 implies

$$p_D^{follow} \geq p_S^{follow},$$

we have

$$\Pi_D^{S \to D} = \lambda p_D^{follow} \geq \lambda p_S^{follow} = \Pi_D^{D \to S},$$

and

$$\Pi_S^{S \to D} = (1-\lambda) p_D^{follow} \geq (1-\lambda) p_S^{follow} = \Pi_S^{D \to S}.$$

Thus, among the matched sequential outcomes, both firms weakly prefer the small-leads/dominant-follows allocation, $S \to D$.

Mechanism. The role preference for $S \to D$ comes from the undercutting incentives that make the higher matched price sustainable. Let $\Delta_i^U(p)$ denote firm i's net gain from undercutting a rival price p, relative to matching. For the dominant firm,

$$\Delta_D^U(p) = (1-\lambda)(p-\delta) - \lambda\delta = (1-\lambda)p - \delta.$$

For the small firm,

$$\Delta_S^U(p) = \lambda(p-\delta) - (1-\lambda)\delta = \lambda p - \delta.$$

The relevant region is where these incentives have opposite signs:

$$\Delta_D^U(p) < 0 < \Delta_S^U(p) \quad \Longleftrightarrow \quad \frac{\delta}{\lambda} < p < \frac{\delta}{1-\lambda}.$$

I refer to this sign reversal as behavioral separation. For prices in this interval, the dominant firm is willing to match, while the small firm would rather undercut. These matched prices are therefore sustainable only when the dominant firm follows. This is the source of role preference alignment: $S \to D$ makes the higher matched price sustainable, and both firms benefit from that higher price.

2.6 A Three-Revision Alternating Illustration

The Stackelberg game clears the market once, after the follower responds. The leader's initial price therefore generates no separate payoff before the follower responds. In posted-price markets, by contrast, a new price takes effect immediately and remains in force until the next revision. I capture this feature by allowing the market to clear after every revision. Each action then determines current profit and sets the price to which the rival subsequently responds. A firm may therefore sacrifice current demand to induce a more favorable rival price at the next revision.

Three revisions are the shortest horizon that captures this tradeoff. The initial mover chooses a price and receives demand, the rival responds and receives demand at the new price pair, and the initial mover then responds in turn. I compare the orders

$$S \longrightarrow D \longrightarrow S \qquad \text{and} \qquad D \longrightarrow S \longrightarrow D.$$

Unlike the Stackelberg game, there is no fixed leader or follower: the firm that responds in one revision becomes the firm awaiting a response in the next. Comparing the two orders shows how demand asymmetry interacts with the identity of the current and future responder. The timing follows the alternating-move logic of Maskin and Tirole (1988), but truncates the interaction after three revisions so that the dynamic incentives remain transparent.

Let i denote the first and third mover, let j denote the second mover, and let x be j's inherited posted price. The first mover chooses a_1, the second mover observes a_1 and chooses a_2, and the first mover then chooses a_3. Let $\beta \in (0, 1]$ weight profits earned in later market-clearing intervals. Firm k's payoff is

$$U_k = \pi_k(\mathbf{p}^1) + \beta\pi_k(\mathbf{p}^2) + \beta^2\pi_k(\mathbf{p}^3),$$

where $\mathbf{p}^\tau$ is the posted-price profile after revision τ. The terminal action is

$$a_3^*(a_2) \in \arg\max_{a \in \mathcal{P}} \pi_i(a, a_2). \tag{2}$$

Thus, the last mover solves the static match-versus-undercut problem characterized by condition (1). Anticipating this terminal response, the second mover chooses

$$a_2^*(a_1) \in \arg\max_{a \in \mathcal{P}} \left\{ \pi_j(a, a_1) + \beta \pi_j\big(a, a_3^*(a)\big) \right\}. \tag{3}$$

The first mover then solves

$$\begin{aligned} a_1^*(x) \in \arg\max_{a \in \mathcal{P}} \Big\{ & \pi_i(a, x) + \beta \pi_i\big(a, a_2^*(a)\big) \\ & + \beta^2 \pi_i\big(a_3^*(a_2^*(a)), a_2^*(a)\big) \Big\}. \end{aligned} \tag{4}$$

Unlike the one-shot leader, the first mover may accept a low current payoff if its action induces a more favorable rival price before its own return move.

To illustrate the mechanism, consider the parameterization

$$\mathcal{P} = \{0.2, 0.4, 0.6, 0.8, 1\}, \qquad \lambda = 0.7, \qquad \beta = 0.99.$$

Backward induction yields a unique path for every inherited price state. Since the first mover immediately replaces its own inherited price, the equilibrium path depends only on the inherited price of the rival. Table 1 reports these paths. Appendix Figure A.1 reports the optimal action at each revision for every initial price pair.

Table 1. Equilibrium Paths in the Three-Revision Illustration

Move order	Inherited rival price	Equilibrium action path
$S \to D \to S$	$p_D^0 \in \{0.2, 0.4\}$	$1.0 \to 0.8 \to 0.6$
	$p_D^0 \in \{0.6, 0.8\}$	$0.6 \to 0.6 \to 0.4$
	$p_D^0 = 1.0$	$0.8 \to 0.6 \to 0.4$
$D \to S \to D$	$p_S^0 \in \{0.2, 0.4, 0.6\}$	$0.2 \to 0.6 \to 0.6$
	$p_S^0 = 0.8$	$0.8 \to 0.6 \to 0.6$
	$p_S^0 = 1.0$	$1.0 \to 0.8 \to 0.6$

Notes: Entries report the actions chosen at the first, second, and third revisions. The market clears after each action. The solution uses $\mathcal{P} = \{0.2, 0.4, 0.6, 0.8, 1\}$, $\lambda = 0.7$, and $\beta = 0.99$.

Table 2. Anticipation and Small-Firm Price Initiation

Small firm's position	Undercuts	Initiates increase	Matched if initiated
Moves last: no subsequent response ($S \to D \to \mathbf{S}$)	5/5	0/5	–
Moves second: dominant responds next ($D \to \mathbf{S} \to D$)	2/5	3/5	3/3

Notes: The table compares the small firm's action across the five economically distinct inherited rival prices. "Initiates increase" means that the small firm posts a price strictly above the dominant firm's current price. The last column conditions on the small firm having initiated such an increase.

Table 2 isolates the mechanism. The dominant firm's anticipated response changes the small firm's initiation decision. When the small firm moves last, it has no future response to influence and undercuts in all five states. When it moves before the dominant firm, it initiates a price increase in three of five states. The dominant firm subsequently matches all three increases. Thus, the comparison holds fixed the small firm but changes whether a dominant response lies ahead.

The path $0.2 \to 0.6 \to 0.6$ gives the clearest example. After the dominant firm chooses 0.2, the small firm raises its price to 0.6 and receives no demand at the second clearing. This temporary sacrifice is optimal because the small firm anticipates that the dominant firm will match 0.6 at the final revision. Thus, anticipation of a dominant response changes small-firm initiation: the prospect of accommodation can induce the small firm to start a price increase that it would not choose as the terminal mover.

Dominant followership affects prices before the dominant firm acts. A small firm that would otherwise undercut may initiate an increase when it expects the dominant firm to match. Thus, predictable accommodation not only sustains a rival's increase but also encourages that increase to occur. Observed price leadership can therefore be misleading: the small firm posts the increase, but the dominant firm's response makes it profitable. The finite horizon identifies this anticipation effect, but not its infinite-horizon persistence.

2.7 Robustness: Loyal Demand Away from Price Parity

The baseline model assumes that a firm charging above its rival receives zero demand. This assumption isolates the matching-versus-undercutting mechanism, but it is stark in many retail settings, where some consumers remain attached to a firm even when it is not the lowest-priced firm. I therefore consider a loyal-switcher extension in which firms retain loyal consumers away from price parity.

Environment. Total demand is again normalized to one. Demand now consists of three groups: a mass α loyal to the dominant firm, a mass β loyal to the small firm, and a mass

$$s = 1 - \alpha - \beta$$

of switchers, where

$$0 < \beta < \alpha, \qquad \alpha + \beta < 1.$$

Loyal consumers buy from their preferred firm regardless of relative prices. Switchers buy from the lower-price firm; if prices are equal, the dominant firm receives share $\lambda \in [1/2, 1]$ of switchers.

Demand for the dominant firm is therefore

$$q_D(p_D, p_S) = \begin{cases} \alpha + s, & p_D < p_S, \\ \alpha + \lambda s, & p_D = p_S, \\ \alpha, & p_D > p_S, \end{cases}$$

while demand for the small firm is

$$q_S(p_D, p_S) = \begin{cases} \beta, & p_D < p_S, \\ \beta + (1 - \lambda)s, & p_D = p_S, \\ \beta + s, & p_D > p_S. \end{cases}$$

Behavioral separation. I use this extension to test whether the behavioral separation identified above can arise when firms retain loyal consumers away from price parity. This is not automatic, because loyal demand changes both the value of matching and the value of deviating from a matched price.

Write the leader price as $p = k\delta$. For each follower, let π_i^{match}, π_i^{under}, and π_i^{high} denote the payoffs from matching the leader, undercutting by one grid step, and taking the loyal-only high-price option. If the dominant firm follows, these payoffs are

$$\pi_D^{match} = k\delta[\alpha + \lambda s], \qquad \pi_D^{under} = (k-1)\delta(1-\beta), \qquad \pi_D^{high} = M\delta\alpha.$$

If the small firm follows, they are

$$\pi_S^{match} = k\delta[\beta + (1-\lambda)s], \qquad \pi_S^{under} = (k-1)\delta(1-\alpha), \qquad \pi_S^{high} = M\delta\beta.$$

The desired separation requires the dominant follower to match,

$$\pi_D^{match} > \max\{\pi_D^{under}, \pi_D^{high}\},$$

which is equivalent to

$$\lambda > \max\left\{\frac{(k-1)(1-\beta) - k\alpha}{ks}, \frac{\alpha(M-k)}{ks}\right\}.$$

The two terms correspond to the dominant follower's two possible deviations: undercutting the leader and relying only on loyal consumers at the high price.

For the desired separation to hold, the small follower must prefer undercutting,

$$\pi_S^{under} > \max\{\pi_S^{match}, \pi_S^{high}\}.$$

This is equivalent to

$$\lambda > \frac{1-\alpha}{ks}$$

and

$$(k-1)(1-\alpha) > M\beta.$$

The first inequality makes undercutting better than matching for the small follower; the second makes undercutting better than the loyal-only high-price option.

Combining these restrictions, behavioral separation holds whenever

$$\lambda > \bar{\lambda}(\alpha, \beta, k, M) \equiv \max\left\{\frac{1}{2}, \frac{(k-1)(1-\beta) - k\alpha}{ks}, \frac{\alpha(M-k)}{ks}, \frac{1-\alpha}{ks}\right\},$$

with the additional condition

$$(k-1)(1-\alpha) > M\beta.$$

Thus, for fixed (α, β, k, M), an admissible $\lambda \in [1/2, 1)$ exists whenever $\bar{\lambda}(\alpha, \beta, k, M) < 1$ and the additional condition holds.

Figure 2 illustrates this region for $M = 5$. Each panel fixes the leader price $p = k\delta$. The color gives the minimum parity-switcher share $\bar{\lambda}(\alpha, \beta, k, M)$ required for behavioral separation; gray regions indicate primitives for which no $\lambda < 1$ satisfies the conditions. Darker regions therefore correspond to more robust separation, because the mechanism works with a weaker parity advantage.

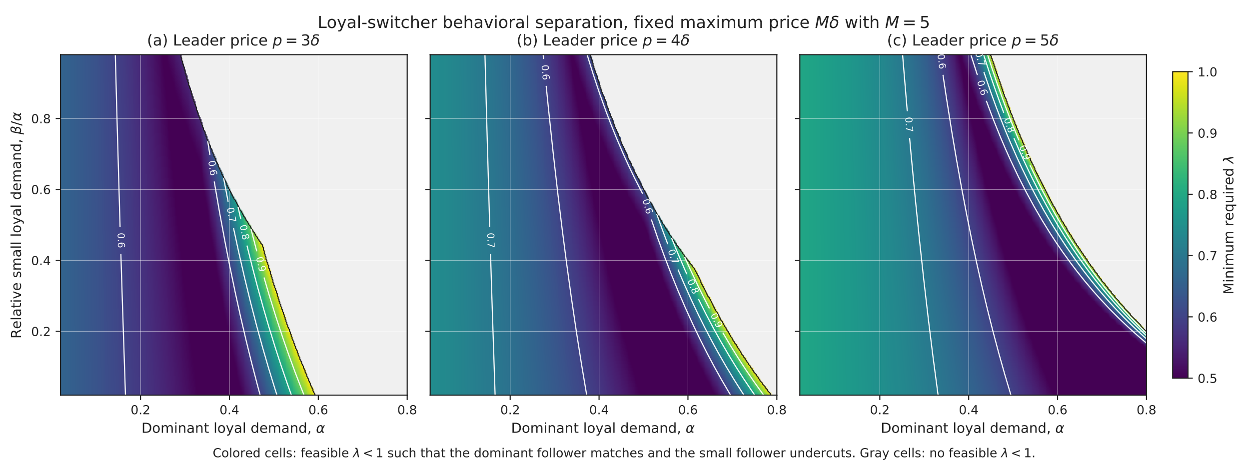

Figure 2. Loyal-Switcher Behavioral Separation. The figure plots the minimum λ required for the dominant follower to match while the small follower undercuts. The maximum price is fixed at $M\delta$ with $M = 5$. Colored regions satisfy $\bar{\lambda}(\alpha, \beta, k, M) < 1$; gray regions admit no feasible $\lambda < 1$.

The colored region shows where the same behavioral separation from the baseline survives after allowing loyal demand away from price parity. The restriction is substantive because loyal consumers affect both sides of the follower's tradeoff. A firm's own loyal demand makes matching more attractive, while the rival's loyal demand limits the demand that can be gained by undercutting. Separation therefore requires a balance: the dominant firm must have enough demand at the matched price to make matching attractive, but enough demand must remain contestable for the small follower to gain from undercutting.

Figure 3 summarizes how demanding this requirement is. For each threshold λ^*, the curve reports the share of the $(\alpha, \beta/\alpha)$ grid for which behavioral separation can be generated with some admissible $\lambda \leq \lambda^*$.

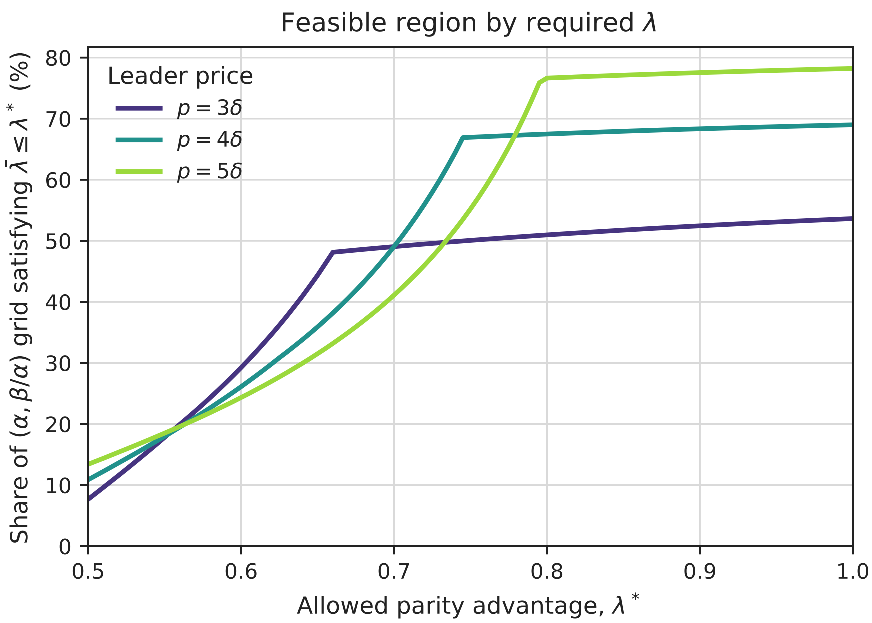

Figure 3. Feasible Region by Required Demand Advantage. For each leader price $p = k\delta$, the curve reports the share of loyal-demand primitives for which behavioral separation holds with required $\lambda \leq \lambda^*$. Curves that rise earlier correspond to regions that require a weaker parity advantage.

The takeaway is that the loyal-switcher environment admits a nonempty region in which the dominant-followership mechanism survives. For those primitives, some $\lambda \in [1/2, 1)$ makes the dominant firm willing to match as the follower while the small firm prefers to undercut. In the plotted example, the curves begin to rise well below one, so the mechanism is not confined to knife-edge cases with λ arbitrarily close to one. At the same time, the additional mass that appears at higher price levels often requires larger λ, showing that higher prices can expand the feasible set while also making the required parity advantage more demanding.

3 Data and Descriptive Evidence

3.1 Data Overview

I study cross-platform price competition in a real online market by focusing on Amazon's price position and its responses to rival platforms offering the same product. This requires observing not only price levels, but also the timing of price changes across platforms selling identical products. The timing dimension is central to recent algorithmic-pricing debates, including the FTC's Project Nessie allegations, because competitive outcomes may depend on whether dominant platforms respond systematically to rivals' price changes.

To make price comparisons within exact-product markets, I construct a high-frequency panel of matched products across online platforms. The sample is designed around two criteria: products are high-ranking within their Amazon categories, and the same product is also sold by at least one non-Amazon platform. Products are matched at the exact SKU, model specification, and color, so comparisons hold the underlying product fixed. For each product, I follow every online platform on which I observe the product for sale. The platform set therefore varies by product and includes online superstores, brick-and-mortar retailers' online stores, branded stores, and smaller specialized platforms. [3]

The sample covers electronics and home appliances from June 2024 to June 2025. Narrow product categories include headphones, earbuds, speakers, smart-home devices, wearables, computing products, storage devices, vacuum and floor-care products, home goods, and pet supplies. Prices were collected daily early in the sample and every two hours after high-frequency collection began on November 24, 2024.

Table 3 summarizes the panel. The data contain 216 matched products, 829 product-platform pairs, and 1.20 million in-stock price observations. The average product is observed on 3.84 platforms. Amazon is observed for nearly all matched products.

[3] The FTC uses a related distinction between online superstores, which offer broad product depth and breadth, and more specialized online stores in its complaint against Amazon (Federal Trade Commission 2024).

Table 3. Overview of Matched Cross-Platform Pricing Data

Metric	Value
Sample period	Jun. 2024–Jun. 2025
Matched products	216
Product-platform pairs	829
Avg. platforms per product	3.84
Median platforms per product	4.00
Avg. scrape interval after Nov. 24, 2024	2.2 hours
Price observations	1,198,066

Notes: A matched product is an identical SKU/specification/color combination tracked across platforms. Price observations exclude out-of-stock listings and missing prices. The scraping interval is computed after the high-frequency collection began on November 24, 2024.

3.2 Price Competitiveness and Matching

The theory highlights the dominant firm's willingness to match a rival's price rather than cut below it. When the dominant firm follows, this willingness to match can make higher rival prices sustainable; when this response channel is absent, price competition is disciplined by the incentive to cut below rivals. This mechanism requires the dominant platform to remain close to the market minimum without always pricing strictly below rivals. I therefore first document Amazon's position in the cross-platform price distribution and then ask whether its lowest-price episodes come from unique lowest prices or ties.

Amazon is persistently price competitive across matched products. Each comparison holds the product fixed and compares prices across platforms at the same time. Amazon has the lowest observed price in 78.7% of competitive product-time comparisons. Even when Amazon is not the lowest-priced platform, it remains near the low end of the price distribution: its average relative rank is 0.115, where 0 denotes always lowest and 1 always highest. Figure 4 shows that this pattern is not concentrated in a single category or period.

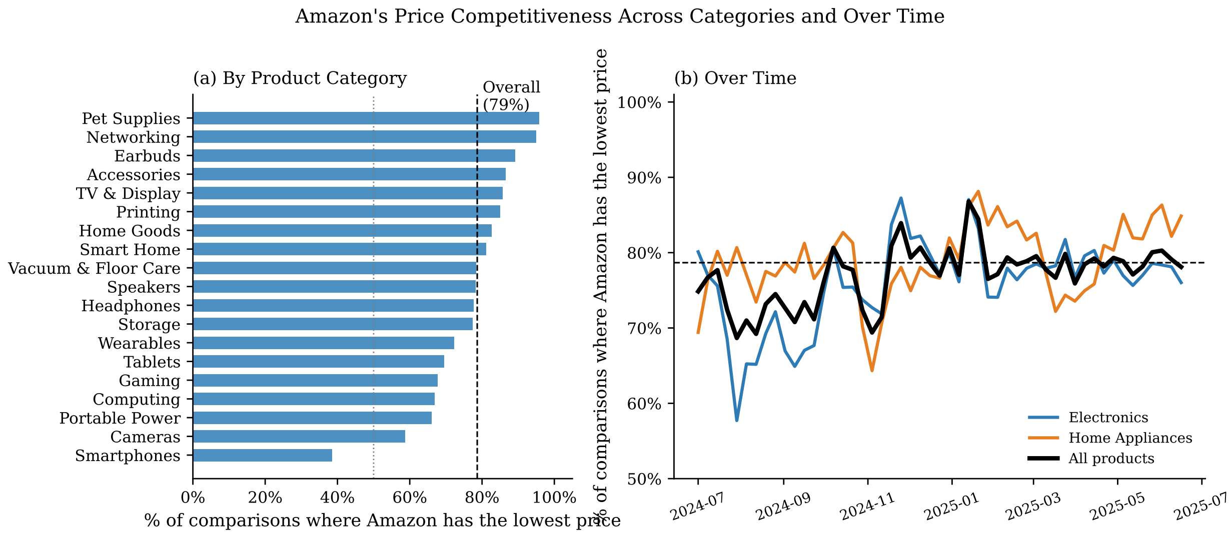

Figure 4. Amazon's Price Competitiveness Across Product Categories and Over Time. Panel (a) reports the fraction of competitive product-time comparisons in which Amazon has the lowest observed price, by product category. Panel (b) plots the same measure over time. Each observation is a product × 2-hour bin in which both Amazon and at least one competitor have a listed price.

Matching accounts for a quantitatively important share of Amazon's low-price position. I split Amazon-lowest observations into cases in which Amazon is uniquely lowest and cases in which Amazon ties with at least one rival at the minimum observed price. Figure 5 shows that ties account for a substantial share of Amazon-lowest observations and exceed unique-lowest episodes in most groups. The pattern is especially stable across demand groups: when products are split into thirds by the Amazon demand proxy, the tie share exceeds 50% in each group. Thus, Amazon's low-price position is not driven only by strict undercutting. This is the empirical counterpart of the model's matching channel: Amazon can remain price competitive while avoiding the margin loss associated with pricing below rivals.

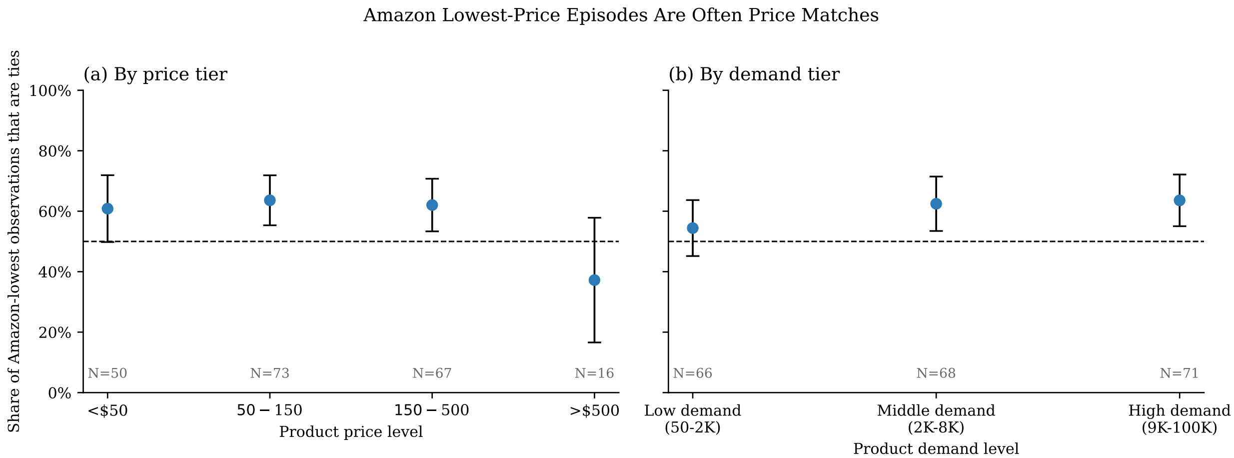

Figure 5. Share of Amazon Lowest-Price Episodes That Are Ties. The figure reports the share of Amazon-lowest product × 2-hour observations in which Amazon is tied with at least one rival at the minimum observed price. Panel (a) groups products by median observed market price. Panel (b) groups products into thirds of each product's median Amazon "Bought in Past Month" proxy. Vertical bars report 95% confidence intervals clustered at the product level.

3.3 Repricing Frequency and Granularity

The matching evidence raises a second question: whether Amazon's pricing process generates enough adjustment activity for response patterns to be visible in the data. If price changes are rare or occur only through large resets, rivals have few opportunities to learn whether their own price changes are followed by subsequent Amazon adjustments. Frequent and relatively granular repricing creates more such opportunities, making it possible both for rivals to form expectations about Amazon's pricing behavior.

Figure 6 shows this pattern in the data. Adjustment frequency is measured as the average number of price changes per product-week, using platform-product observations with a valid prior observation between one and six hours earlier. Amazon averages 2.10 changes per product-week, compared with 0.99 for Walmart and less than one for the remaining focal platforms. Conditional adjustment sizes are modest despite this high frequency: Amazon's mean price increase is 16.5%, close to Walmart's 15.5% and far below 31.8% for Best Buy, 34.9% for Home Depot, and 34.6% for Target. Thus, Amazon reprices more frequently than rival platforms, while its adjustment magnitudes remain relatively granular. Appendix Table A.1 reports the corresponding estimates and product-level standard errors, and Appendix Figure A.3 shows adjustment-size distributions by product price level.

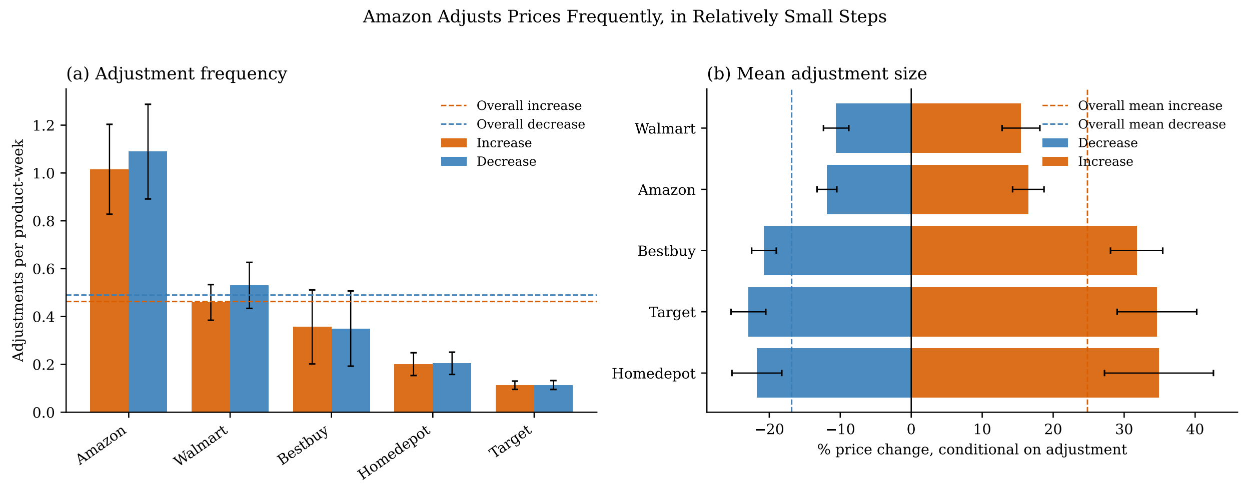

Figure 6. PRICE ADJUSTMENT FREQUENCY AND SIZE BY PLATFORM. Panel (a) reports the average number of price increases and decreases per product-week. Panel (b) reports the mean percentage size of price changes, conditional on an adjustment. Bars report platform-level averages; vertical and horizontal whiskers report 95% confidence intervals computed across products. Dashed lines report the corresponding overall means across platforms and products.

3.4 Sequential Timing of Price Changes

The evidence on frequent repricing raises a timing question: do platforms tend to adjust at the same time, or does one platform's price change typically occur before another platform adjusts?[4] This distinction matters because a response channel requires one price change to be observed before a later adjustment. I therefore measure whether price changes are primarily simultaneous or sequential, and whether an initial price change is followed by another platform's adjustment for the same product within a short response window.

The high-frequency panel measures staggered timing at the two-hour level. Figure 7 classifies product-time price-change bins by the number of platforms that adjust in the same bin. Because prices are observed in discrete two-hour bins, multiple changes within the same bin are treated as unordered same-bin moves. Under this classification, 91.3% of price-change bins involve exactly one eligible platform changing price.

The same figure shows that many one-platform moves are followed by later price changes for the

[4] For the timing, response, and state-dependence analyses below, Amazon observations are restricted to first-party listings sold by Amazon, so observed Amazon prices are Amazon's own retail prices rather than prices set by third-party marketplace sellers.

same product. A 48-hour window captures follow-on price changes for 33.9% of sequential moves. These follow-on adjustments occur soon enough after the initial move to be treated as potential responses, rather than only as unrelated low-frequency repricing. I use this window below to test whether rival price changes predict subsequent focal-platform adjustments.

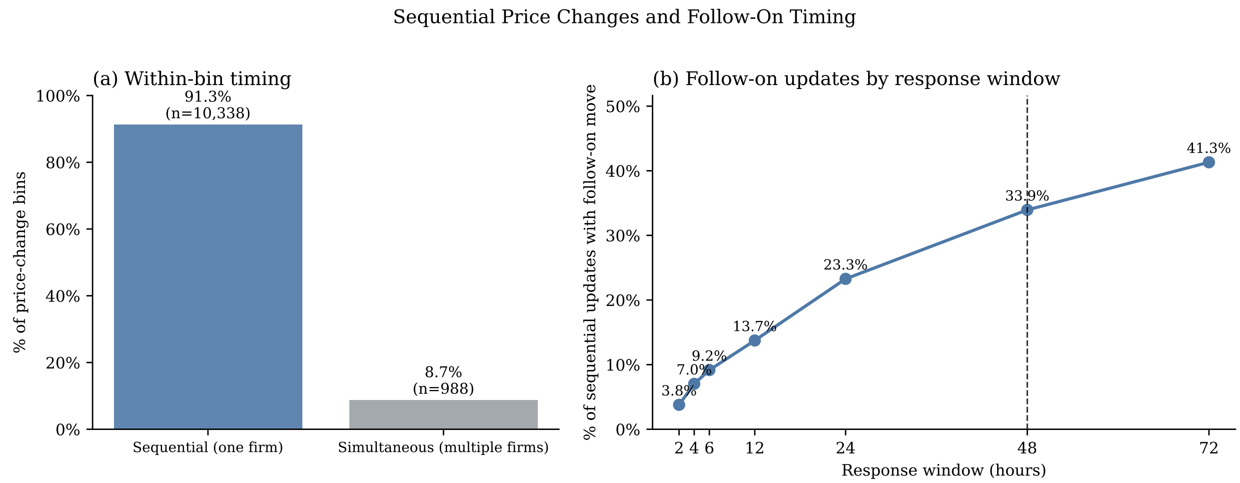

Figure 7. Sequential Price Changes and Follow-On Timing. Panel (a) classifies product-time price-change bins as sequential if exactly one eligible platform changes price and as same-bin multi-platform moves if multiple eligible platforms change price in the same bin. Because prices are observed in two-hour bins, same-bin multi-platform moves are unordered within the bin. Panel (b) reports, among sequential one-platform updates, the cumulative share followed by another platform's price update for the same product within each response window. The dashed vertical line marks the 48-hour response window used below. Timing analyses restrict Amazon observations to first-party listings and identify price updates only when the same platform-product was observed within the prior one to six hours.

3.5 Rival Price Changes and Follow-On Matching

The evidence on sequential timing raises the central response question: when one platform changes price first, which platforms tend to adjust afterward? The empirical test compares periods in which a rival has just moved with periods in which no rival has moved, among cases where the focal platform does not adjust. If the focal platform is mainly an initiator, rival moves should not systematically predict its later price changes. If it is a responder, a rival move should raise its probability of adjusting within the response window. I estimate this relationship first for any rival price change, then separately for rival increases and rival decreases. Finally, for Amazon adjustments that follow same-direction rival moves, I ask whether Amazon's new price remains at parity with or below the relevant rival benchmark.

The first test estimates whether a rival move predicts a later focal-platform adjustment. The unit of observation is platform-product-time. The estimation sample includes only observations in which focal platform i does not change price at time t, so the outcome captures subsequent adjustment rather than contemporaneous movement. The baseline specification is

$$Adjust_{ip,t:t+48} = \sum_{f \in \mathcal{F}} \beta_f \mathbf{1}\{i = f\} \times RivalChange_{ip,t} + \alpha_p + \lambda_w + \eta_d + \delta_i + \varepsilon_{ipt}, \tag{5}$$

where $Adjust_{ip,t:t+48}$ equals one if platform i adjusts the price of product p at least once during $(t, t+48]$, and $RivalChange_{ip,t}$ equals one if at least one rival platform selling the same product changes price at t. The coefficient β_f measures, in percentage points, platform f's excess probability of adjusting after a rival move relative to otherwise comparable periods without a rival move.

All response regressions include product fixed effects α_p, week fixed effects λ_w, day-of-week fixed effects η_d, and platform fixed effects δ_i. Product fixed effects allow some products to be repriced more often than others. Platform fixed effects allow Amazon and the other platforms to have different baseline repricing rates. Week and day-of-week fixed effects absorb broad shocks and systematic repricing rhythms, such as promotional cycles or market-wide cost changes. The coefficients therefore ask whether a rival move predicts additional adjustment beyond a product's typical repricing frequency, a platform's baseline adjustment rate, and common calendar variation. Standard errors are clustered by product.

Panel (a) of Figure 8 shows that Amazon is the most responsive major platform. After a rival price change, Amazon's probability of adjusting within 48 hours rises by 7.2 percentage points. The corresponding estimates for the other major platforms are small and statistically indistinguishable from zero.

I next ask whether this response is concentrated in upward moves. The direction-specific specifications are

$$Increase_{ip,t:t+48} = \sum_{f \in \mathcal{F}} \beta_f^+ \mathbf{1}\{i = f\} \times RivalIncrease_{ip,t} + \alpha_p + \lambda_w + \eta_d + \delta_i + \varepsilon_{ipt}, \tag{6}$$

$$Decrease_{ip,t:t+48} = \sum_{f \in \mathcal{F}} \beta_f^- \mathbf{1}\{i = f\} \times RivalDecrease_{ip,t} + \alpha_p + \lambda_w + \eta_d + \delta_i + \varepsilon_{ipt}. \tag{7}$$

The first equation uses focal-platform price increases after rival increases; the second uses focal-platform price decreases after rival decreases. Thus, β_f^+ and β_f^- measure same-direction response to rival increases and decreases, respectively.

Panel (b) shows that Amazon's response is strongest after rival price increases. After a rival increase, Amazon is 12.0 percentage points more likely to raise its own price within 48 hours. The estimates for the other major platforms are much smaller, and Amazon's response to rival increases is larger than its response to rival decreases. Table 4 reports the corresponding estimates. Appendix Figure A.2 reports grouped robustness specifications using 12-, 24-, 48-, and 72-hour response windows.

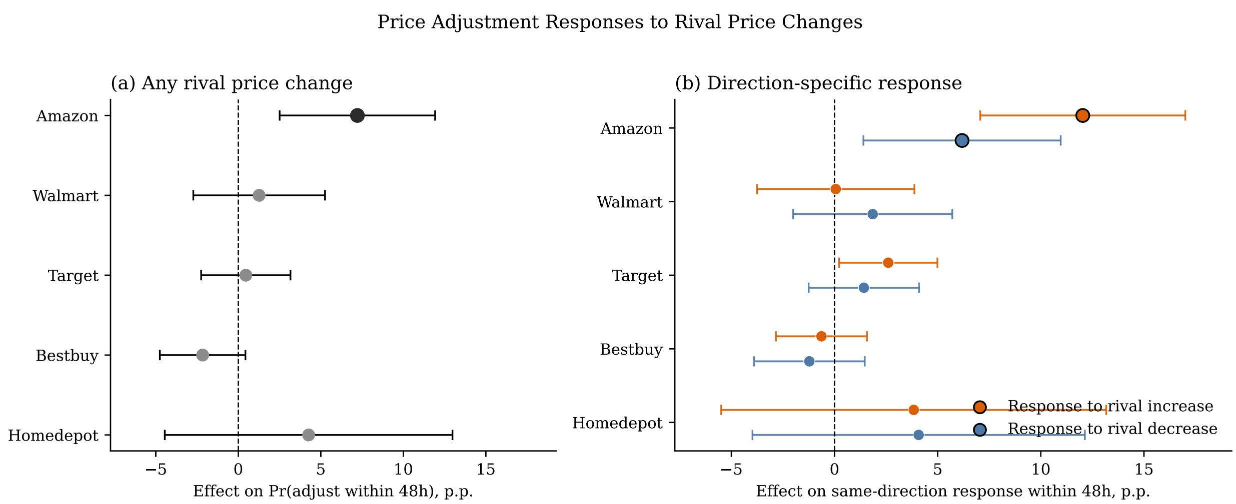

Figure 8. Response to Rival Price Changes. The figure plots percentage-point effects from linear probability models of focal-platform price adjustment within 48 hours of a rival price change. Panel (a) uses any rival price change as the treatment. Panel (b) separates rival price increases and decreases and uses same-direction focal-platform adjustment as the outcome. All specifications include product, week, day-of-week, and platform fixed effects. Standard errors are clustered by product.

The final step asks what Amazon does when it follows. Instead of classifying all Amazon-lowest

Table 4. Price Adjustment Responses to Rival Price Changes

	(1) Any adjustment	(2) Price increase	(3) Price decrease
Amazon × Rival event	7.22***	12.03***	6.18**
	(2.40)	(2.54)	(2.44)
Walmart × Rival event	1.26	0.06	1.85
	(2.04)	(1.94)	(1.97)
Target × Rival event	0.46	2.61**	1.42
	(1.38)	(1.21)	(1.37)
Bestbuy × Rival event	-2.16	-0.63	-1.22
	(1.32)	(1.13)	(1.37)
Homedepot × Rival event	4.26	3.84	4.08
	(4.45)	(4.76)	(4.11)
Product FE	Yes	Yes	Yes
Week FE	Yes	Yes	Yes
Day-of-week FE	Yes	Yes	Yes
Platform FE	Yes	Yes	Yes
Clustered SE	Product	Product	Product
Observations	678,357	678,357	678,357

Notes: The unit of observation is platform-product-time. The focal platform is required not to change price at t. Column (1) estimates whether any rival price change at t predicts a focal-platform price change in $(t, t+48]$. Columns (2) and (3) estimate direction-specific responses. Coefficients are percentage-point effects from linear probability models with product, week, day-of-week, and platform fixed effects. Standard errors, clustered by product, are in parentheses.

observations, I condition on Amazon adjusting after a same-direction rival move and ask where Amazon lands. For each Amazon adjustment, I look backward 48 hours for prior rival moves in the same direction and compare Amazon's post-adjustment price to the lowest current price among those prior-moving rivals.

Figure 9 shows that Amazon's follow-on adjustments usually preserve a competitive price position. After rival increases, Amazon remains below or at parity with the benchmark in 81.7% of same-direction follow-on adjustments; after rival decreases, Amazon is at parity or below in 83.3%. Thus, Amazon's response is not simply more frequent repricing. Conditional on moving after rivals, Amazon usually moves to a price that matches or remains below the relevant rival benchmark.

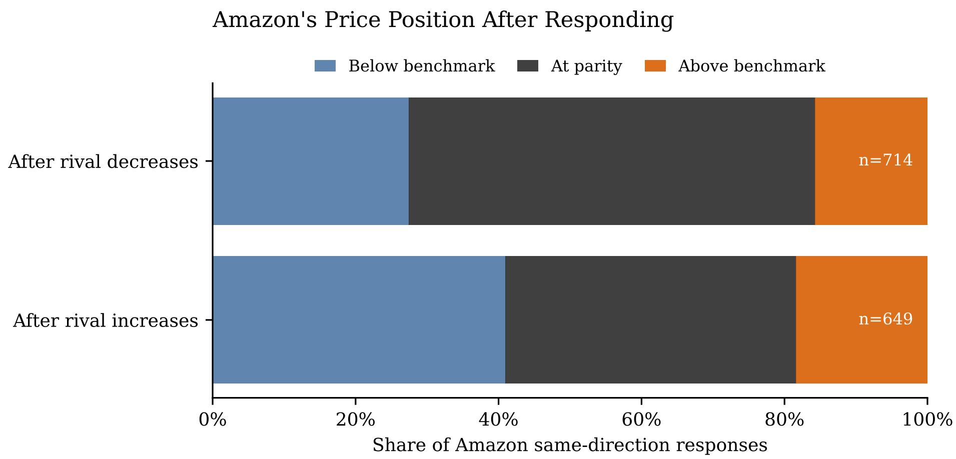

Figure 9. Amazon's Price Position After Responding to Rival Moves. The figure classifies Amazon price adjustments that follow same-direction rival moves within the prior 48 hours. The benchmark is the lowest current price among prior-moving rivals at Amazon's adjustment time. "At parity" allows a tolerance of the larger of $1 or 1% of the benchmark price.

Together, these results identify the response pattern behind the matching evidence. Rival price changes predict Amazon's later adjustments; the response is especially strong after rival increases; and when Amazon follows same-direction rival moves, it usually lands at parity with or below the rivals that moved first. This is the empirical pattern that other platforms could learn from repeated interaction: Amazon is not merely often low-priced, but responds to rival moves in a way that maintains its competitive price position.

3.6 State Dependence in Prices

The response evidence suggests that platforms condition on rivals' prices, but the dynamic model also requires a parsimonious state representation. I therefore ask whether current prices are summarized by two lagged objects: the platform's own price and the lowest rival price. This distinction matters for the followership mechanism because a platform's relevant choice is not an absolute price in isolation, but a price relative to the rival benchmark it may match, undercut, or exceed.

The unit of analysis is a platform-product observation with a valid prior observation for the same platform and product. Let $p_{ip,t}$ denote platform i's price for product p at observed time t, and define the lowest rival price as

$$p^{\min}_{-i,p,t} = \min_{j \neq i} p_{jp,t}.$$

The lag $t-1$ denotes the previous observed state for the same platform-product pair, restricted to observations between one and six hours apart.

I use two transformations to interpret price adjustment. The first is the change in the focal platform's price and the change in the lowest rival price:

$$\Delta \log p_{ip,t} = \log p_{ip,t} - \log p_{ip,t-1}, \qquad \Delta \log p^{\min}_{-i,p,t-1} = \log p^{\min}_{-i,p,t-1} - \log p^{\min}_{-i,p,t-2}.$$

The second is the focal platform's lagged position relative to the lowest rival price:

$$Gap_{ip,t-1} = \log p_{ip,t-1} - \log p^{\min}_{-i,p,t-1}.$$

I estimate three specifications. The first measures own-price persistence. The second asks whether the lagged lowest rival price helps explain the focal platform's current price, conditional on its own lagged price. The third asks whether platforms move toward the rival benchmark:

$$\log p_{ip,t} = \rho \log p_{ip,t-1} + \alpha_p + \lambda_w + \eta_d + \delta_i + \varepsilon_{ipt}, \tag{8}$$

$$\log p_{ip,t} = \rho \log p_{ip,t-1} + \theta \log p^{\min}_{-i,p,t-1} + \alpha_p + \lambda_w + \eta_d + \delta_i + \varepsilon_{ipt}, \tag{9}$$

$$\Delta \log p_{ip,t} = \theta \Delta \log p^{\min}_{-i,p,t-1} + \kappa Gap_{ip,t-1} + \alpha_p + \lambda_w + \eta_d + \delta_i + \varepsilon_{ipt}. \tag{10}$$

All specifications include product fixed effects, week fixed effects, day-of-week fixed effects, and platform fixed effects; standard errors are clustered by product.

Table 5. State-Space Motivation: Own and Rival Price States

	(1) $\log p_{ip,t}$	(2) $\log p_{ip,t}$	(3) $\Delta \log p_{ip,t}$
$\log p_{ip,t-1}$	0.9812***	0.9793***	
	(0.0039)	(0.0046)	
$\log p^{\min}_{-i,p,t-1}$		0.0056**	
		(0.0022)	
$\Delta \log p^{\min}_{-i,p,t-1}$			0.0093
			(0.0067)
$Gap_{ip,t-1}$			-0.0136***
			(0.0036)
Product FE	Yes	Yes	Yes
Week FE	Yes	Yes	Yes
Day-of-week FE	Yes	Yes	Yes
Platform FE	Yes	Yes	Yes
Clustered SE	Product	Product	Product
Observations	969,928	969,928	938,258
R^2	0.999	0.999	0.007

Notes: The unit of observation is platform-product-time. The lag $t-1$ is the previous observed state for the same platform-product pair, restricted to observations between 1 and 6 hours apart. $p^{\min}_{-i,p,t}$ is the lowest rival price in the same product market, Δ denotes log price changes, and $Gap_{ip,t-1} = \log p_{ip,t-1} - \log p^{\min}_{-i,p,t-1}$. All specifications include product, week, day-of-week, and platform fixed effects. Column (3) requires two consecutive valid lags. Standard errors, clustered by product, are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The estimates support using both own and rival prices in the dynamic state. Column (1) shows strong own-price persistence: a platform's current price is closely related to its previous price. Column (2) shows that own-price persistence is not sufficient by itself: the lagged lowest rival price helps explain the focal platform's current price even after conditioning on the focal platform's own lagged price. These estimates motivate a dynamic state representation that includes both a platform's own lagged price and the lagged lowest rival price.

Column (3) shows that the relative position of these two prices predicts the direction of adjustment. The coefficient on $Gap_{ip,t-1}$ is negative and statistically significant: platforms priced above the lowest rival subsequently adjust downward, while platforms priced below the lowest rival tend to adjust upward. Figure 10 shows the same pattern nonparametrically.

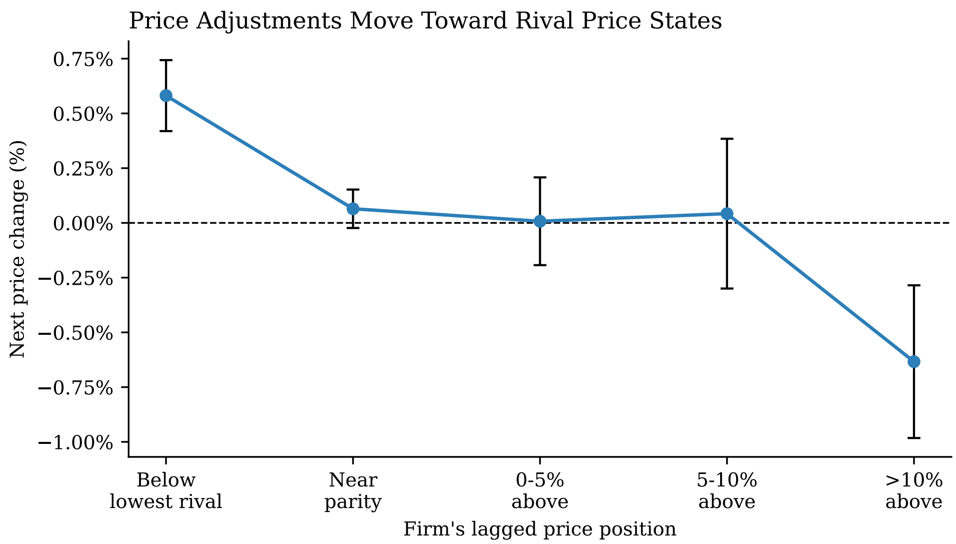

Figure 10. Relative Price Position and Subsequent Adjustment. The figure bins platform-product-time observations by the implied lagged gap, $\log p_{ip,t-1} - \log p^{\min}_{-i,p,t-1}$, and plots the average next log price change, $\Delta \log p_{ip,t}$. Each point first averages within product-bin cells and then averages across products; confidence intervals use variation across products. Platforms below the rival benchmark tend to raise prices, while platforms far above it tend to cut prices.

4 AI-Driven Pricing Dynamics

The empirical evidence has three implications for the model. First, price changes are not simultaneous: one platform often changes price before another platform adjusts. Second, rival price changes predict Amazon's later adjustments, especially same-direction responses after rival increases. Third, when Amazon adjusts after same-direction rival moves, it usually remains at parity with or below the rivals that moved first.

The theoretical model explains why this pattern can matter for prices. A demand-advantaged firm gives up more margin when it undercuts, so it may prefer to match a rival's price. This force raises prices in the theoretical model only when the dominant firm moves second and can respond to the rival's posted price.

The remaining question is whether the same force survives repeated learning. In online retail, platforms do not choose prices once in a known payoff environment. Instead, they repeatedly interact, observe rivals' posted prices, update their decisions, and learn from realized profits over time. To examine whether the static intuition persists in this dynamic setting, I maintain the same demand system and discrete price grid as before, but allow firms to learn pricing strategies through reinforcement learning, specifically Q-learning.

4.1 Environment

Time is discrete and indexed by t. There are two firms, indexed by $i \in \{1, 2\}$, and $-i$ denotes firm i's rival. Firms choose prices from the finite grid $\mathcal{P}$, so the action space is $\mathcal{A} = \mathcal{P}$. Let $a_{i,t}$ denote the price chosen when firm i moves, and let $p_{i,t}$ denote its posted price after period-t updating.

When firm i is called to move, it observes the posted price profile inherited from the previous period. The baseline firm-specific state is

$$s_{i,t} = (p_{i,t-1}, p_{-i,t-1}).$$

Thus, the state space is $\mathcal{S} = \mathcal{P} \times \mathcal{P}$. A pricing policy for firm i is a map $\sigma_i : \mathcal{S} \to \mathcal{P}$. This state specification follows the empirical evidence that both a firm's own lagged price and the lagged rival benchmark help explain subsequent prices.

After the updating protocol determines the posted price profile $p_t = (p_{1,t}, p_{2,t})$, firm i receives period profit

$$r_{i,t} = \pi_i(p_{i,t}, p_{-i,t}),$$

which serves as the reward in the learning problem.

I next describe the price-updating protocols used to generate this repeated interaction.

4.2 Timing Protocols and Learning

The timing protocol carries the static followership mechanism into a repeated posted-price environment. In the theoretical model, the leader chooses a price, the follower observes it, and payoffs are realized after the follower responds. In online retail, however, prices remain posted until a platform changes them, so a price change can affect demand before the rival responds and remains part of the state observed at the rival's next adjustment opportunity.

The simultaneous benchmark removes this response channel: firms choose prices at the same time, without observing each other's current choice, and demand is realized only after both prices are set. I therefore use an alternating protocol that preserves the key feature of followership, responding to a rival's posted price, while embedding it in a repeated learning environment.

Simultaneous updating. In the simultaneous benchmark, both firms move in every period. Given $s_{i,t} = (p_{i,t-1}, p_{-i,t-1})$, each firm i chooses an action $a_{i,t} \in \mathcal{P}$. The posted price profile is

$$p_t = (p_{1,t}, p_{2,t}) = (a_{1,t}, a_{2,t}).$$

Demand is realized at p_t, and firm i receives

$$r_{i,t} = \pi_i(p_{i,t}, p_{-i,t}).$$

Firm i's next state is

$$s_{i,t+1} = (p_{i,t}, p_{-i,t}).$$

Alternating updating with persistent prices. In the alternating protocol, only one firm moves in each period. Suppose firm i is active in period t, so firm $-i$ is inactive. Firm i observes $s_{i,t} = (p_{i,t-1}, p_{-i,t-1})$ and chooses $a_{i,t} \in \mathcal{P}$. Prices evolve according to

$$p_{i,t} = a_{i,t}, \qquad p_{-i,t} = p_{-i,t-1}.$$

Demand is realized at the resulting price profile, and firms receive

$$r_{i,t} = \pi_i(p_{i,t}, p_{-i,t}), \qquad r_{-i,t} = \pi_{-i}(p_{-i,t}, p_{i,t}).$$

The next decision state for firm $-i$ is

$$s_{-i,t+1} = (p_{-i,t}, p_{i,t}).$$

Thus, firm i's posted price remains in effect when firm $-i$ becomes active in period $t+1$.

Learning. Under both protocols, firms choose actions using Q-learning whenever they are called to move. For each state s and price a, $Q_{i,t}(s, a)$ denotes firm i's current estimate of the discounted continuation value of choosing a in state s. Firms use these estimates to choose prices according to an ϵ_t-greedy policy, where ϵ_t is the exploration probability. At each decision epoch, firm i chooses a greedy action

$$a_{i,t} \in \arg\max_{a \in \mathcal{P}} Q_{i,t}(s_{i,t}, a)$$

with probability $1 - \epsilon_t$, and draws uniformly from $\mathcal{P}$ with probability ϵ_t.

The Q-learning update combines the current value estimate with a newly realized target, using learning rate α_t:

$$Q_i(s_{i,t}, a_{i,t}) \leftarrow (1 - \alpha_t) Q_i(s_{i,t}, a_{i,t}) + \alpha_t \widehat{Q}_{i,t}(s_{i,t}, a_{i,t}),$$

where $\widehat{Q}_{i,t}$ is the payoff evidence generated after firm i chooses action $a_{i,t}$ in state $s_{i,t}$. The two timing protocols differ in this target.

The state $s_{i,t}$ indexes the Q-value being updated in both protocols. The protocols differ in how the components of this state enter the target. In the simultaneous benchmark, both firms update their Q-functions after each period. After actions are chosen, the realized target is

$$\widehat{Q}^{sim}_{i,t}(s_{i,t}, a_{i,t}) = r_{i,t} + \beta \max_{a' \in \mathcal{P}} Q_{i,t}(s_{i,t+1}, a').$$

At the moment of choice, however, firm i has not observed the rival's current action. Conditional on state and action, the corresponding target is therefore an expectation over the rival's simultaneous choice:

$$\mathbb{E}\left[\pi_i(a_{i,t}, a_{-i,t}) + \beta \max_{a' \in \mathcal{P}} Q_{i,t}(s_{i,t+1}, a') \mid s_{i,t}, a_{i,t}\right].$$

Thus, the rival-price component of the state, $p_{-i,t-1}$, affects the simultaneous target only indirectly: it helps firm i predict the distribution of $a_{-i,t}$ and the next state generated by the simultaneous price profile.

In the alternating protocol, a firm's decision epoch spans two periods. Suppose firm i is active in period t. Its price remains in effect when firm $-i$ becomes active in period $t + 1$. After firm $-i$ updates, the state at firm i's next decision epoch is

$$s_{i,t+2} = (p_{i,t+1}, p_{-i,t+1}) = (p_{i,t}, p_{-i,t+1}),$$

where the last equality uses the fact that firm i is inactive in period $t+1$. The target is therefore the two-period return

$$\widehat{Q}_{i,t}^{alt}(s_{i,t}, a_{i,t}) = r_i(a_{i,t}, p_{-i,t-1}) + \beta r_i(a_{i,t}, p_{-i,t+1}) + \beta^2 \max_{a' \in \mathcal{P}} Q_{i,t}(s_{i,t+2}, a'),$$

where $p_{-i,t+1}$ is the rival's subsequent price choice. Here $p_{-i,t-1}$ enters the target in two distinct ways. First, it is the rival's currently posted price, so it enters the immediate payoff directly through $r_i(a_{i,t}, p_{-i,t-1})$. Second, it is part of the state from which the rival responds, since

$$s_{-i,t+1} = (p_{-i,t-1}, a_{i,t}).$$

It can therefore affect the rival's response $p_{-i,t+1} = \sigma_{-i}(s_{-i,t+1})$. That response then enters firm i's target through the second-period reward $r_i(a_{i,t}, p_{-i,t+1})$ and through the next decision state $s_{i,t+2} = (a_{i,t}, p_{-i,t+1})$.

The distinction is the source of the price-response channel. In the simultaneous benchmark, lagged rival prices help firm i predict the distribution of $a_{-i,t}$, but the firm does not observe a currently posted rival price when choosing $a_{i,t}$. In the alternating protocol, the same state component is the rival's price currently in force. Firm i's action is therefore a response to an observed posted price, affects demand immediately, and becomes the price observed by the rival at the next adjustment opportunity. The comparison therefore separates two forces. Both protocols allow firms to learn from lagged prices. Only the alternating protocol lets a firm actually follow a rival's posted price and observe the payoff consequence of that response. Symmetrically, the firm that moves first observes the consequence of initiating a price change that remains in force when the rival next moves. Holding the demand system, action space, state variables, and learning rule fixed, the alternating protocol isolates whether these realized consequences of sequential price responses can sustain the dominant-followership mechanism in a repeated environment.

Learning parameters. The discount factor is fixed at $\beta = 0.99$. The learning and exploration rates follow the same hyperbolic decay schedule over a simulation horizon of T periods:

$$\alpha_t = \max\left\{\underline{\alpha}, \alpha_0 \frac{\kappa_\alpha}{\kappa_\alpha + t}\right\}, \qquad \epsilon_t = \max\left\{\underline{\epsilon}, \epsilon_0 \frac{\kappa_\epsilon}{\kappa_\epsilon + t}\right\}.$$

In the baseline simulations, $\alpha_0 = \epsilon_0 = 0.1$ and $\underline{\alpha} = \underline{\epsilon} = 0.05$. I set $\kappa_\alpha = \kappa_\epsilon = 0.1T$, so that learning and exploration decline gradually over the simulation horizon but remain bounded away from zero.

4.3 Simulation Design

Main counterfactual: fixed initiator-response roles. Dominant followership is a statement about who responds to whom. A dynamic simulation must therefore distinguish a price change that initiates a market response from a price change that follows one. This distinction is central to the mechanism: when firms differ in demand advantage, the price consequences of a response may depend on whether the responder is the dominant firm or the small firm. The simulation makes this role assignment explicit by placing firms in fixed initiator and responder roles.

The main counterfactual fixes the asymmetric demand environment from Section 2 and varies only which firm occupies the response role. Let D denote the dominant firm and S the small firm. The comparison has two cases: D as the responder and S as the responder. It asks whether the same demand advantage has different price effects depending on whether the dominant firm or the small firm responds.

The fixed initiator-response protocol modifies the alternating protocol described above. The alternating protocol provides the natural starting point: prices are persistent, and firms move sequentially. By itself, however, it cannot isolate dominant followership, because each firm alternates between initiating a price change and responding to the rival's previous move. The design therefore imposes two additional restrictions that make the responder role persistent and economically interpretable.

First, adjustment opportunities are tied to role. The initiator receives the opportunity to revise its posted price. The responder observes the initiator's posted price and revises only if the initiator changed price. If the initiator's price is unchanged, the responder's posted price remains fixed.

Second, responder actions are local. When the responder revises, it chooses among three responses to the initiator's posted price: undercut by one grid point, match, or price one grid point above. These restrictions focus the simulation on the response margin documented in the empirical analysis.

Formally, let L denote the initiator and F the responder. At initiator periods, L chooses

$p_{L,t} \in \mathcal{P}$. At responder periods, F observes the initiator's posted price. If $p_{L,t} \neq p_{L,t-2}$, then

$$p_{F,t} \in \{p_{L,t} - \Delta,\ p_{L,t},\ p_{L,t} + \Delta\} \cap \mathcal{P},$$

where Δ is one grid step. If $p_{L,t} = p_{L,t-2}$, then

$$p_{F,t} = p_{F,t-1}.$$

Thus, responder actions are tied to observed initiator price changes rather than to an independent revision clock. The dominant-responder case is the dynamic analogue of dominant followership: the dominant firm observes an initiated price change, chooses a local response, and learns from the realized payoff consequences of that response.

Benchmark 1: simultaneous updating. The first benchmark is the canonical simultaneous-learning environment. It compares symmetric and asymmetric demand when firms choose current prices at the same time. Firms can learn from lagged prices, but they cannot respond to a currently posted rival price. This benchmark identifies what demand asymmetry does in the absence of the posted-price response channel.

Benchmark 2: unrestricted alternating roles. The second benchmark is the unrestricted alternating protocol described above. It adds the posted-price response channel: only one firm revises in a period, and the rival's price remains posted until its next opportunity to move. The comparison is again symmetric demand versus asymmetric demand, now under alternating updating. Here firms can respond to posted rival prices, but no firm is permanently assigned to the response role: the identity of the responder alternates over time, and both firms choose from the full price grid. This benchmark identifies what demand asymmetry does when posted-price responses are possible, but before imposing a fixed dominant-responder role. The fixed initiator-response design then adds that final ingredient by assigning the response role to either the dominant firm or the small firm.

Benchmark 3: adjustment speed. The third benchmark varies the speed of response within the posted-price environment. It starts from the unrestricted alternating protocol, but delays one firm's next adjustment opportunity. When the delayed firm's turn arrives, the market clears for additional periods at the existing posted prices before that firm can revise. Firms earn profits during these delay periods, and the Q-update includes the full reward path between a firm's current action and its next decision epoch. For an action whose rival response is delayed by one additional market-clearing period, the target is

$$\widehat{Q}_{i,t}^{speed} = r_i(a_{i,t}, p_{-i,t-1}) + \beta r_i(a_{i,t}, p_{-i,t-1}) + \beta^2 r_i(a_{i,t}, p_{-i,t+1}) + \beta^3 \max_{a' \in \mathcal{P}} Q_{i,t}(s_{i,t+3}, a').$$

The first two reward terms are earned at the same posted price profile because the delayed firm has not yet revised; the third reward is earned after the delayed firm chooses $p_{-i,t+1}$. The comparison is between the unrestricted alternating benchmark and an otherwise identical environment with a one-period response delay. This benchmark isolates the role of response timing within the posted-price environment, holding fixed the alternating structure of price revision.

4.4 Results

Demand asymmetry under simultaneous updating. Figure 11 compares symmetric and asymmetric demand under simultaneous updating, isolating the effect of demand asymmetry when firms cannot respond to a rival's current posted price. In the final 10 percent of simulation periods, the average minimum market price is 0.338 under symmetric demand and 0.274 under asymmetric demand; the difference is not statistically different from zero at the 5 percent level. Thus, absent the posted-price response channel, demand asymmetry does not raise the competitive price in the market.

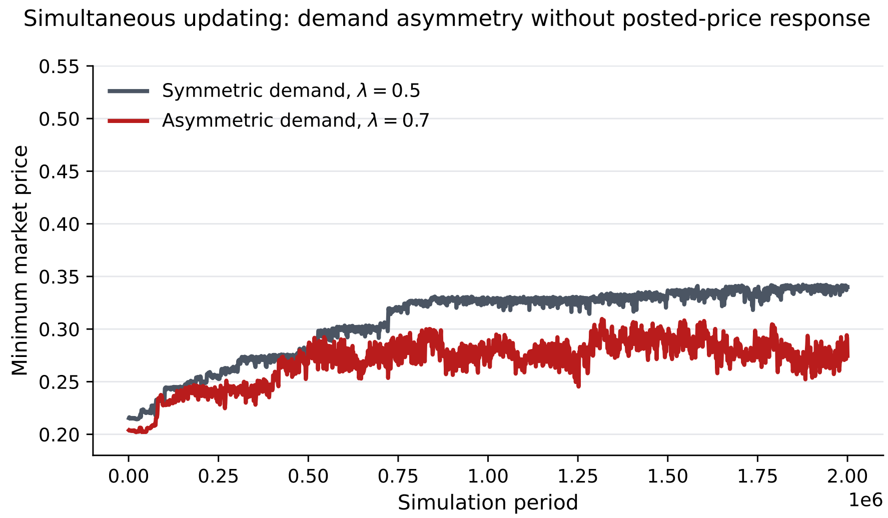

Figure 11. Demand Asymmetry under Simultaneous Updating. The figure reports the mean minimum market price across simulation runs under simultaneous updating. Within each run, prices are averaged over non-overlapping 1,000-period bins. The plotted series averages these bin-level prices across 30 runs.

Posted-price response without fixed roles. Figure 12 compares simultaneous and alternating updating under symmetric and asymmetric demand, isolating the effect of posted-price response when no firm is fixed in the responder role. Holding the demand environment fixed, alternating updating raises the minimum market price in both panels. In the final 10 percent of simulation periods, the average minimum market price rises from 0.338 to 0.571 under symmetric demand and from 0.274 to 0.464 under asymmetric demand. Among the alternating simulations, however, asymmetric demand does not produce higher prices than symmetric demand. Thus, allowing firms to respond to posted rival prices raises prices, but demand asymmetry does not raise prices when the responder role alternates between firms.

Figure 12. Posted-Price Response without Fixed Roles. The figure reports the mean minimum market price across simulation runs. Within each run, prices are averaged over non-overlapping 1,000-period bins. The plotted series averages these bin-level prices across 30 runs.

Adjustment speed. Figure 13 compares unrestricted alternating updating with and without a one-period response delay under symmetric demand, isolating the effect of response speed within the posted-price environment. The delay has little effect on prices. In the final 10 percent of decision events, the average minimum market price is 0.571 with no response delay and 0.570 with a one-period delay. Thus, adding a one-period delay in the rival's response does not materially change the long-run market price in this benchmark.

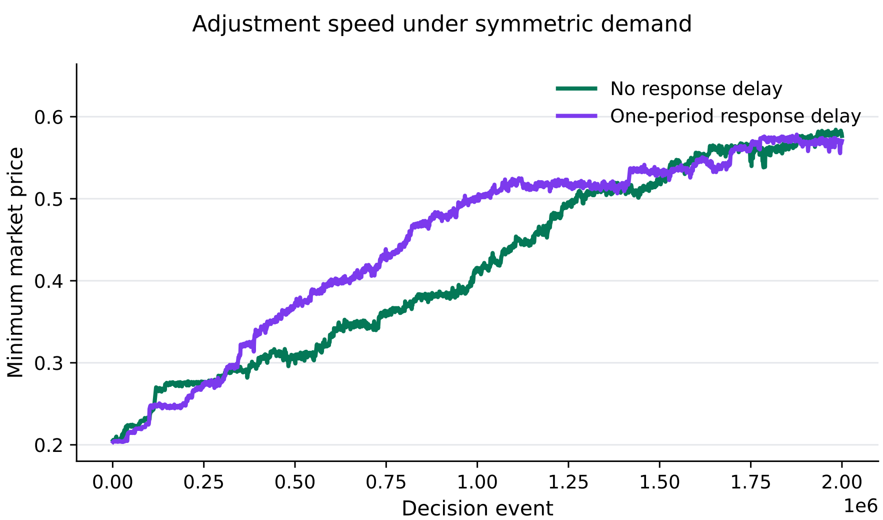

Figure 13. Adjustment Speed. The figure reports the mean minimum market price across simulation runs under symmetric demand. The one-period delay series is plotted in decision-event time: extra market-clearing observations that occur before the delayed firm adjusts are excluded from the plotted path. Within each run, prices are averaged over non-overlapping 1,000-decision-event bins. The plotted series averages these bin-level prices across 30 runs.

Dominant responder. Figure 14 compares the two fixed-role counterfactuals under asymmetric demand, isolating the effect of assigning the dominant firm to the responder role rather than the initiator role. Prices are higher when the dominant firm responds. In the final 10 percent of simulation periods, the average minimum market price is 0.632 when the dominant firm initiates and 0.785 when the dominant firm responds; the difference is statistically different from zero at the 5 percent level. Joint profits also rise in the dominant-responder case, indicating that this role assignment is not only price-elevating but also more profitable for firms jointly. Appendix Figure A.5 shows that the two firms' posted prices remain comparable, so the higher market price is not driven by one firm pricing far above the other. Thus, the fixed-role counterfactual supports the central mechanism: prices are higher, and the configuration is more sustainable, when the demand-advantaged firm is the one that responds.

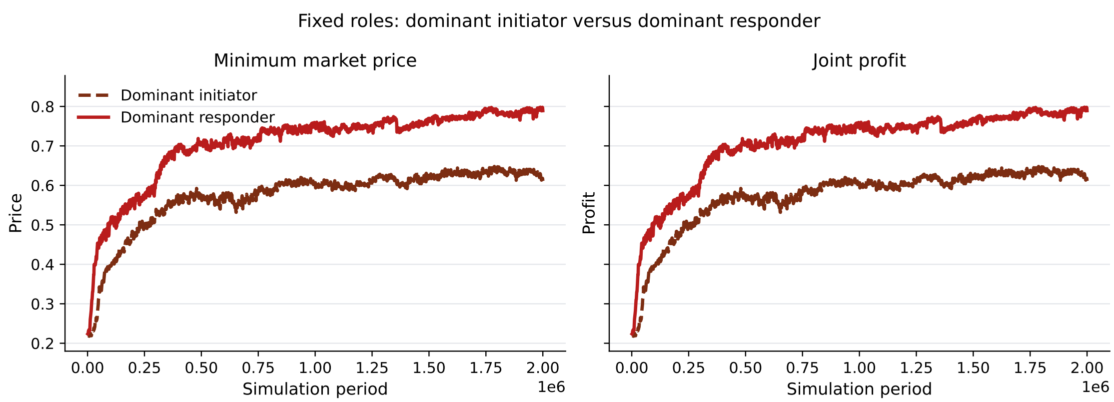

Figure 14. Market Outcomes under Fixed Roles. The figure compares fixed initiator-response roles under asymmetric demand. Within each run, prices and profits are averaged over non-overlapping 1,000-period bins. The plotted series averages these bin-level outcomes across 30 runs.

Figure 15 shows how the learned price path differs across role assignments. In the final 10 percent of simulation periods, the dominant-initiator case is concentrated around intermediate posted-price states. The dominant-responder case instead shifts toward high-price states: the share of states in which both prices are at least 0.8 rises from 0.349 to 0.719. Thus, assigning the dominant firm to the responder role changes the states the algorithms learn to revisit, moving the market toward high-price regions of the state space.

Figure 15. Visited Price States under Fixed Roles. The heat maps report the distribution of posted-price states in the final 10 percent of simulation periods. Rows are dominant-firm prices and columns are small-firm prices. Cell labels report percentages and are shown for cells with shares of at least 2.5 percent.

Figure 16 traces the learned value of matching a rival price rather than undercutting it. For each training decile, I compute $Q(\text{match}) - Q(\text{undercut})$ in each posted-price state and weight those differences by how often the state is visited in that decile. The figure therefore measures the matching incentive in the states that are relevant along the realized learning path. When the dominant firm is the responder, this margin rises sharply for the dominant firm, while the small firm's margin remains negative. By contrast, when the dominant firm initiates, the small firm is the responder and does not learn a positive matching margin. Thus, the high-price states in Figure 15 arise when the responder is the dominant firm: the dominant responder learns to match rather than undercut, whereas the small responder does not.

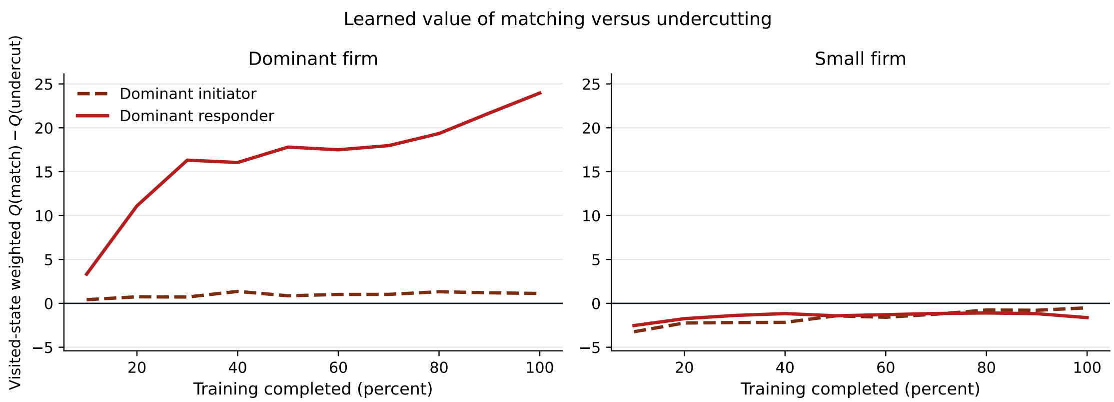

Figure 16. Learned Value of Matching versus Undercutting. The figure reports the visited-state weighted value difference between matching and undercutting the rival's posted price. The weights are the posted-price states visited in the corresponding training decile. Positive values indicate that matching has higher learned value than undercutting.

5 Conclusion

This paper studies how algorithmic responsiveness can interact with demand-side market power. The central mechanism is dominant followership. A demand-advantaged firm has less to gain from undercutting a rival because it already retains a larger share of demand at parity. When that firm moves second, it may therefore prefer to match a rival's price rather than cut below it. This response can make higher rival prices sustainable and raise market prices without explicit coordination.

The empirical evidence documents the ingredients of this mechanism in online retail. Amazon is frequently at the market minimum, and its low-price position often comes from ties rather than unique lowest prices. Price changes are frequent and sequential enough for response patterns to be observed. Rival price changes predict Amazon's later adjustments, especially after rival increases, and Amazon's follow-on adjustments usually leave it at parity with or below the prior-moving rivals. These facts are consistent with a pricing environment in which rivals can learn that Amazon responds to their price changes.

The learning simulations show why this response pattern matters. Demand asymmetry alone does not substantially raise prices when firms move simultaneously. Prices rise when firms move sequentially and the dominant firm is placed in the responder role: learning shifts the market toward states in which matching has high value for the dominant firm, but not for the smaller firm.

These results point to a policy question that is distinct from both explicit collusion and standard monopoly pricing. A monopoly or dominant position is not itself the antitrust concern; the concern is whether that position is used in a way that reduces consumer welfare. Dominant followership raises a hard version of this question because the conduct can look like ordinary competitive matching. The dominant platform need not initiate a price increase or charge more than rivals. Instead, its willingness to match rival increases can make those increases more profitable and sustainable. The relevant object of inquiry is therefore not only the price the dominant firm charges at a point in time, but how its algorithmic response rule changes rivals' incentives. Monitoring response patterns, not only price levels, is central to understanding when AI-driven pricing turns demand-side market power into a market-wide force for higher prices.

A Additional Model Figures

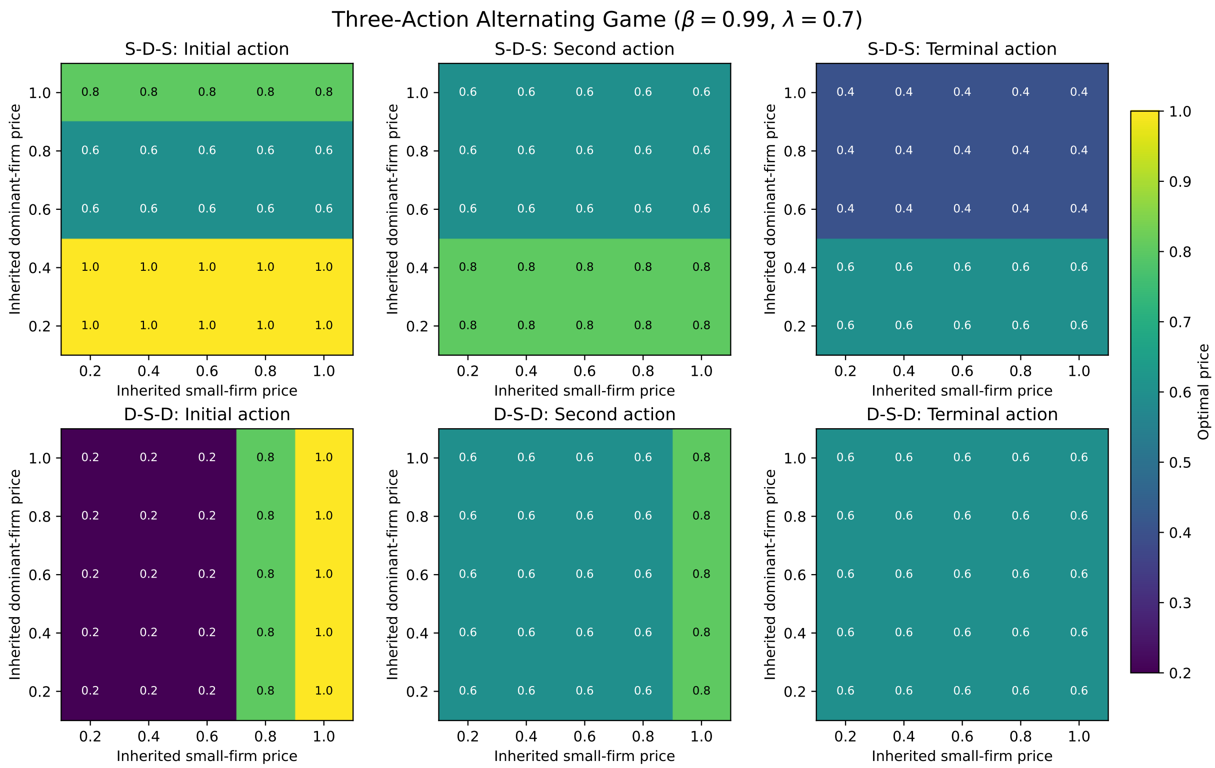

Figure A.1. Optimal Actions in the Three-Revision Illustration.

Notes: The figure reports the backward-induction solution for every initial price pair. The top row shows the $S \to D \to S$ order and the bottom row the $D \to S \to D$ order. Columns report the first, second, and terminal actions. Each cell gives the optimal price. The solution uses $\mathcal{P} = \{0.2, 0.4, 0.6, 0.8, 1\}$, $\lambda = 0.7$, and $\beta = 0.99$. Because the first mover immediately replaces its inherited price, actions vary only with the inherited price of the rival.

B Additional Empirical Tables and Figures

Table A.1. Price Adjustment Patterns by Platform

Platform	Products	Eligible obs.	Any/week	Increase/week	Decrease/week	Mean increase	Mean decrease
Amazon	215	298,100	2.10	1.02	1.09	16.5	11.9
			(0.19)	(0.10)	(0.10)	(1.1)	(0.7)
Walmart	125	153,772	0.99	0.46	0.53	15.5	10.6
			(0.08)	(0.04)	(0.05)	(1.3)	(0.9)
Bestbuy	134	183,823	0.71	0.36	0.35	31.8	20.7
			(0.16)	(0.08)	(0.08)	(1.9)	(0.9)
Homedepot	37	57,705	0.41	0.20	0.20	34.9	21.7
			(0.05)	(0.02)	(0.02)	(3.8)	(1.7)
Target	94	126,597	0.23	0.11	0.11	34.6	22.9
			(0.02)	(0.01)	(0.01)	(2.8)	(1.2)

Notes: The unit is a platform-product-time observation with a valid prior observation for the same platform and product between 1 and 6 hours earlier. Frequency columns report the average number of price adjustments per product-week. Mean magnitudes are conditional on the corresponding type of price change and are reported as percent changes. Displayed platforms are the platforms with the largest number of eligible observations. Standard errors, reported in parentheses, are computed across products.

Table A.2. Amazon Lowest-Price Episodes by Product Price Tier

Price tier	Products	Comparisons	Lowest/tied	Unique lowest	Tied lowest	Ties among lowest
<$50	50	70,250	78.2	30.6	47.6	60.9 [49.8, 71.9]
$50–$150	73	112,765	81.5	29.7	51.9	63.6 [55.3, 71.9]
$150–$500	67	91,620	77.3	29.4	48.0	62.0 [53.3, 70.7]
>$500	16	20,766	70.4	44.2	26.2	37.2 [16.6, 57.8]

Notes: The table decomposes Amazon product-time observations by product price tier, where tiers are based on each product's median observed market price. "Lowest/tied" is the share of comparisons in which Amazon is at the lowest observed price. "Unique lowest" and "Tied lowest" split this share into cases where Amazon is strictly below all rivals versus tied with at least one rival at the minimum price. The final column reports the share of Amazon-lowest observations that are ties. All shares are reported in percentage points; brackets show product-clustered 95 percent confidence intervals.

Table A.3. Amazon Lowest-Price Episodes by Product Demand Tier

Demand tier	Range	Products	Comparisons	Median demand	Lowest/tied	Unique lowest	Tied lowest	Ties among lowest
Low demand	50-2K	66	86,800	400	75.8	34.6	41.3	54.4 [45.2, 63.7]
Middle demand	2K-8K	68	100,091	4,000	76.0	28.5	47.5	62.5 [53.5, 71.5]
High demand	9K-100K	71	107,054	10,000	83.4	30.4	53.0	63.6 [55.0, 72.1]

Notes: Product demand tiers are terciles of each product's median Amazon "Bought in Past Month" proxy. The table decomposes Amazon product-time observations into cases where Amazon is not at the lowest observed price, uniquely lowest, or tied with at least one rival at the minimum price. The final column reports the share of Amazon-lowest observations that are ties. All shares are reported in percentage points; brackets show product-clustered 95 percent confidence intervals.

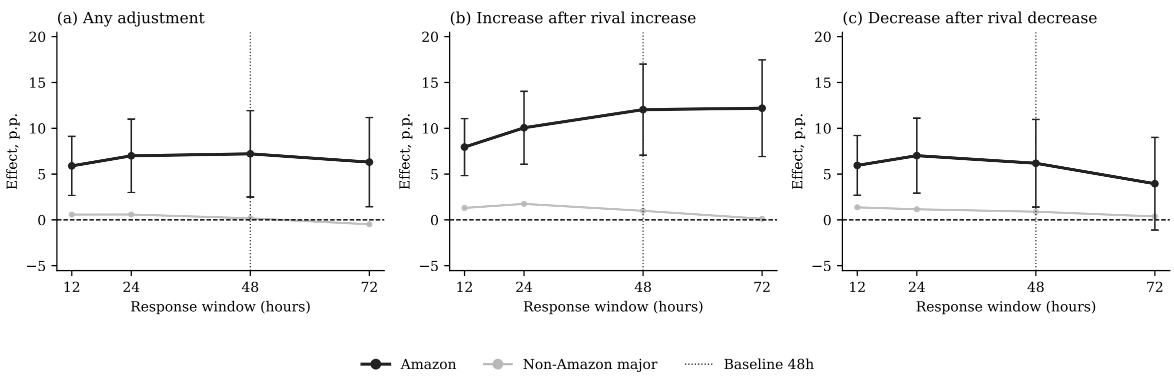

Figure A.2. Response Estimates by Response Window.

Notes: The figure reports percentage-point effects from grouped versions of equations (5)–(7) using alternative response windows. Each specification includes separate treatment interactions for Amazon and for the pooled non-Amazon major platforms; platform fixed effects are retained separately. The vertical dotted line marks the 48-hour window used in the baseline specification. All specifications include product, week, day-of-week, and platform fixed effects. Standard errors are clustered by product.

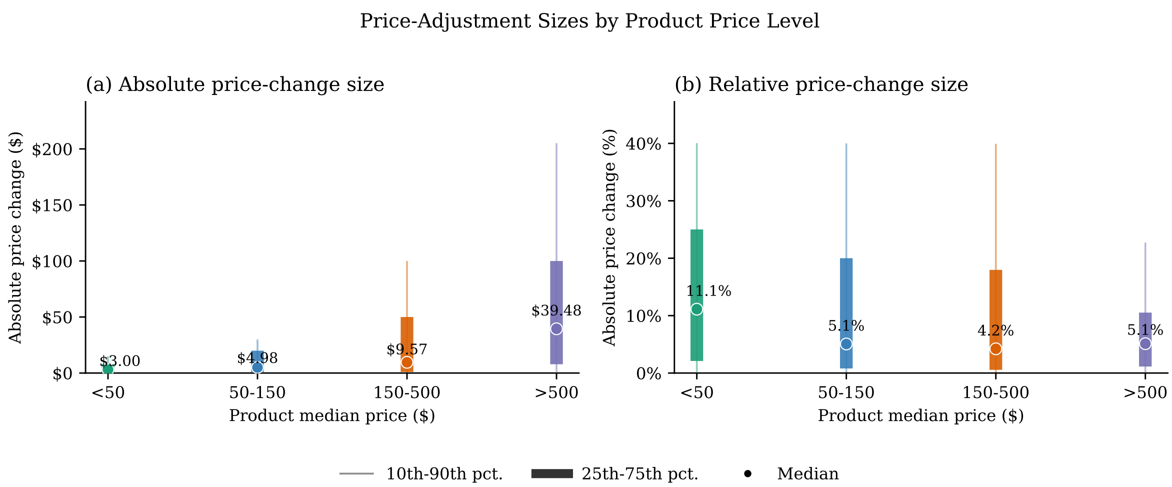

Figure A.3. Price-Adjustment Sizes by Product Price Level.

Notes: The figure reports the distribution of nonzero price-adjustment magnitudes by product median price group. The left panel reports absolute dollar changes, and the right panel reports absolute percentage changes. Thick bars show the interquartile range, thin lines show the 10th–90th percentile range, and circles show medians. Price changes are computed from platform-product observations with a valid prior observation between one and six hours earlier.

C Additional Simulation Figures

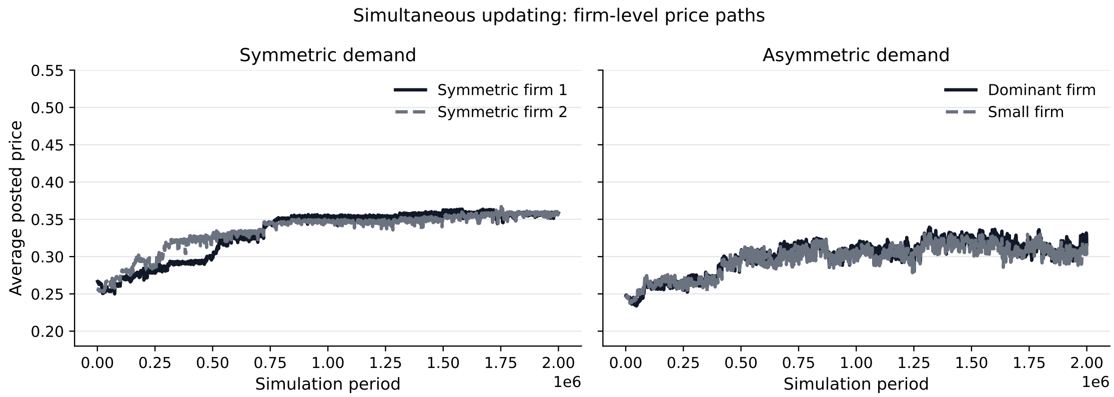

Figure A.4. Firm-Level Prices under Simultaneous Updating. The figure reports mean posted prices by firm under simultaneous updating. Within each run, prices are averaged over non-overlapping 1,000-period bins. The plotted series averages these bin-level prices across 30 runs.

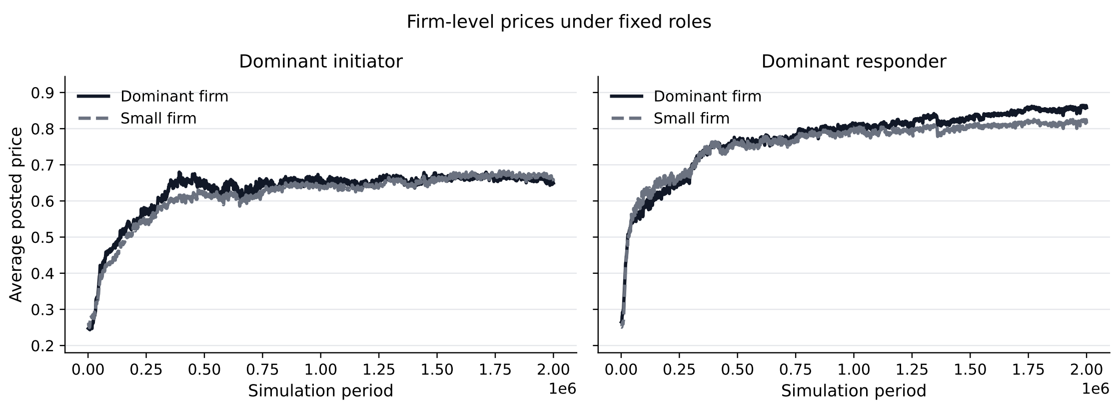

Figure A.5. Firm-Level Prices under Fixed Roles. The figure reports mean posted prices by firm under the two fixed-role counterfactuals. Within each run, prices are averaged over non-overlapping 1,000-period bins. The plotted series averages these bin-level prices across 30 runs.

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